Organic Pollutant Monitor
OPSA-150
For Vietnam

Instruction Manual

CODE:GZ0000369992

Preface

This manual describes the operation of the Organic Pollutant Monitor, OPSA-150.
Be sure to read this manual before using the product to ensure proper and safe operation of the product. Also safely store the manual so it is readily available whenever necessary.

Product specifications and appearance, as well as the contents of this manual are subject to change without notice.

Warranty and responsibility

HORIBA, Ltd. warrants that the Product shall be free from defects in material and workmanship and agrees to repair or replace free of charge, at option of HORIBA, Ltd., any malfunctioned or damaged Product attributable to responsibility of HORIBA, Ltd. for a period of one (1) year from the delivery unless otherwise agreed with a written agreement. In any one of the following cases, none of the warranties set forth herein shall be extended;

- Any malfunction or damage attributable to improper operation
- Any malfunction attributable to repair or modification by any person not authorized by HORIBA, Ltd.
- Any malfunction or damage attributable to the use in an environment not specified in this manual
- Any malfunction or damage attributable to violation of the instructions in this manual or operations in the manner not specified in this manual
- Any malfunction or damage attributable to any cause or causes beyond the reasonable control of HORIBA, Ltd. such as natural disasters
- Any deterioration in appearance attributable to corrosion, rust, and so on
- Replacement of consumables

HORIBA, LTD. SHALL NOT BE LIABLE FOR ANY DAMAGES RESULTING FROM ANY MALFUNCTIONS OF THE PRODUCT, ANY ERASURE OF DATA, OR ANY OTHER USES OF THE PRODUCT.

Trademarks

Company names and brand names are either registered trademarks or trademarks of the respective companies. (R), (TM) symbols may be omitted in this manual.

Regulations

FCC rules

Any changes or modifications not expressly approved by the party responsible for compliance shall void the user's authority to operate the equipment.

Warning

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.
Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Korea certification

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For Your Safety

Hazard classification and warning symbols

Warning messages are described in the following manner. Read the messages and follow the instructions carefully.

- **Hazard classification**

⚠DANGER	This indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. This is to be limited to the most extreme situations.
⚠WARNING	This indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠CAUTION	This indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

- **Warning symbols**

❗	Description of what should be done, or what should be followed
🛇	Description of what should never be done, or what is prohibited

Safety precautions

This section provides precautions for using the product safely and correctly and to prevent injury and damage. The terms of DANGER, WARNING, and CAUTION indicate the degree of imminency and hazardous situation. Read the precautions carefully as it contains important safety messages.

⚠ WARNING
Do not open the cover inside the door of the control unit while the power is ON. Before opening the cover, unplug or turn off the power of the unit to avoid the danger of electric shock.
Do not attempt to disassemble or modify the unit. Doing so could lead to bodily harm and/or injury or may cause damage to the unit or the sample.
Maintain ground to avoid electric shock.
Fire or electric shock • Do not bundle the power supply cord during use. • Do not damage, bend, or stretch the power supply cord forcibly. • If it can not be plugged into an electrical outlet firmly, stop use of the power supply cord. If may result in overheating, fire, or an electrical shock.

⚠ CAUTION
The motor generates a high temperature. When replacing the components of the motor, disassemble the motor after it has cooled down sufficiently.
Take care when handling the cell of the analyzer unit. The cell of the analyzer unit is made of glass. Handle with great care so as not to break the glass.
Do not touch the cell or directly view the UV lamp of the analyzer unit when you check the unit operation.

Product Handling Information

Operational precautions

Use of the product in a manner not specified by the manufacturer may impair the protection provided by the product. And it may also reduce product performance.
Exercise the following precautions:

- Only use the product including accessories for their intended purpose.
- Use the product in the temperature range shown in the general specifications.
- Use the product in an environment free from corrosive gas.
- Avoid using the product in a location close to product using strong electric power (such as an electric furnace), or near a radio or sound wave source.
- Do not give a shock or large vibrations to the unit. To move the product to another location, contact our service or sales office.
- Avoid turning ON the power again immediately after turning it OFF.
- Do not press buttons using wet hands, or sharp objects such as pen tips or screwdrivers.

Disposal of the product

When disposing of the product, follow the related laws and/or regulations of your country.

Manual Information

Description in this manual

Note

This interprets the necessary points for correct operation and notifies the important points for handling the product.

Reference

This indicates the part where to refer for information.

Tip

This indicates reference information.

Documents related to this product

The following documents are related to this product.

Instruction manual for OPSA-150 (this manual)

This volume contains information about necessary work and procedures for daily measurement operations.

Installation manual for OPSA-150

This volume contains information about the installation procedure.

Contents

1 Overview

1.1 Introduction

The OPSA-150 Organic Pollutant Monitor provides continuous measurement of organic pollutants and ultraviolet (253.7 nm) absorption in industrial wastewater, river water, and saltwater. A typical continuous measuring device for drainage water or environmental water cannot provide stable measurement values over long periods of use due to generation of drift from the deteriorated light source/detector or cell contamination from the sample water. In general, once a cell becomes contaminated, the contamination grows gradually until we know it is too serious. To efficiently avoid contamination, continuous cleaning is most important. It is also the key to realize long-term, stable monitoring. The proprietary rotary cell length modulation method ensures stable monitoring. Cleaning is performed using powerful wipers to continuously clean 2 spinning cylinders (cell windows) without obstructing the optical path, thus eliminating read-out errors caused by contamination.

Continual correction for deterioration of the optical system, changes in the sensitivity of the detector, and the effect of contamination on the sample cell give highly consistent and accurate readings.

The analyzer system detects both UV and visible light (546.1 nm) absorbances simultaneously. The TSS correction factor (α) can be selected within the range of 0 to 9.999, according to the sample water.

When this unit is utilized for measuring organic pollutants to determine compliance with water quality gross volume regulations, improved correlation between ultraviolet absorbance and COD values obtained by the designated measuring method is required.
In activated sludge treated sewage water, humic acid, fulvic acid, and tannin, which are the derivative of microbial degradation of humic substances processed by microorganisms, tend to remain. Ultraviolet rays are absorbed by humic acid, fulvic acid and tannin acid. The absorption of ultraviolet rays is often correlated with organic pollution affecting COD. Thus, the correlation between the absorbance of ultraviolet light and organic pollution is improved in many cases.
Also the TSS obtained by VIS is displayed as well as the COD value.

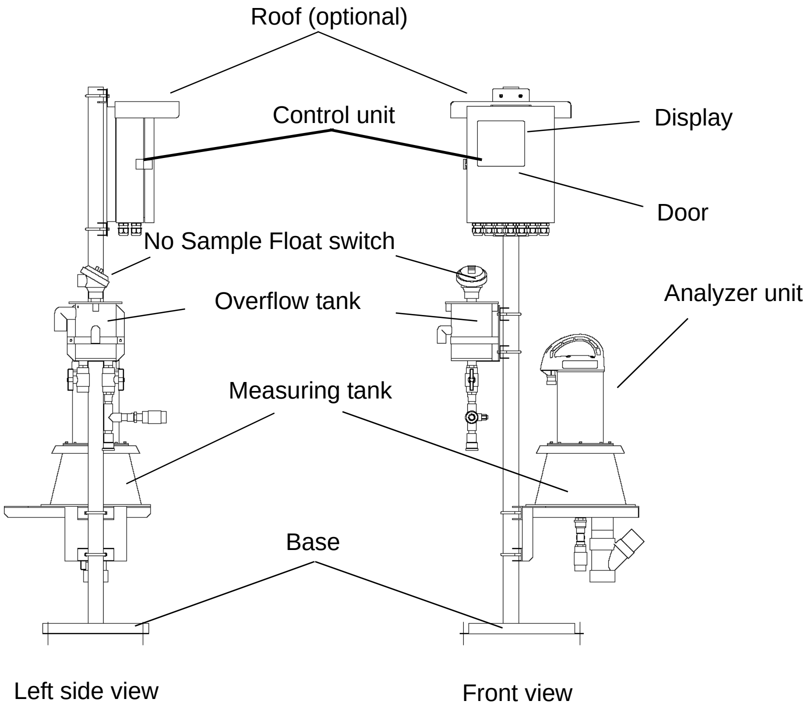
● Features

- Employment of a unique system of rotary cell length modulation in the analyzer unit intrinsically eliminates zero-drift, and a proprietary wiper-method cleaning mechanism provides continuous cleaning, thus allowing stable measurement values over long periods of use.
- The unit features an easy-to-control monochrome graphics LCD data display (touch-panel).
- The display shows UV, VIS (visible light) and UV-αVIS simultaneously and can also switch between converted the COD value and converted TSS displays.
- Measured data can be output to the recorder as analog current output, or transferred to the communication unit (optional) or loaded into a PC via the RS-232C port.
- Measured data can be stored and read using a CF (Compact flash) card.

1.2 Description of Parts

1.2.1 Unit

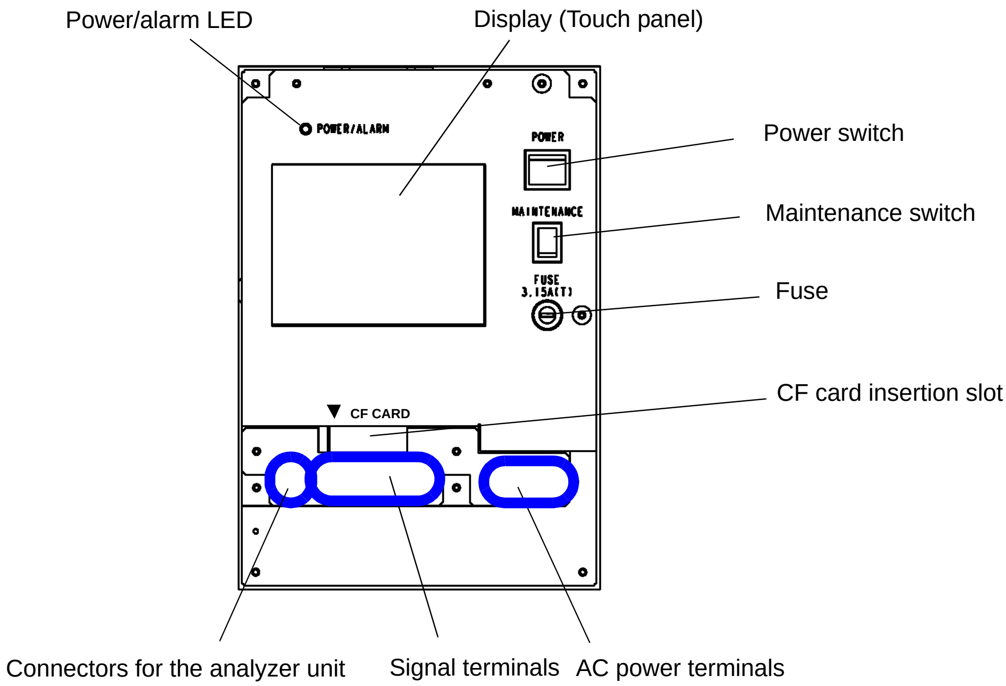
● Front view and side view



Note

The pole stand and the base illustrated are typical examples. Appearance and size may vary depending on the specifications.

● Inside of the control unit



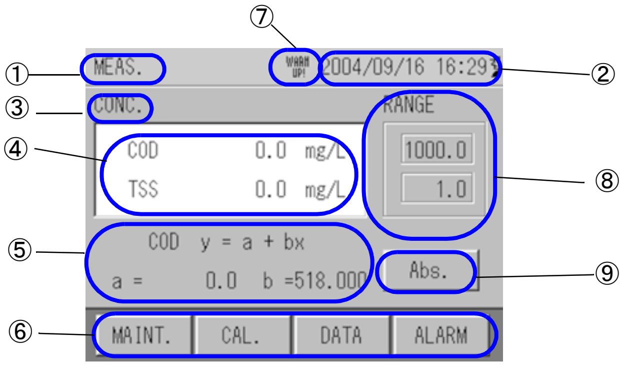
1.2.2 Display

This section describes the parameters shown on a typical display:

- The display is touch panel. Do not press buttons with wet hands, or a sharp object such as a pen tip or a screwdriver.
- If no buttons are pressed for a certain period of time, the backlight turns OFF.
- If no buttons are pressed for about two hours on displays other than those for measurement, data or alarm, the display turns to the measurement display.

● Example of display - Displaying measurement readings

The display shows measurement readings.

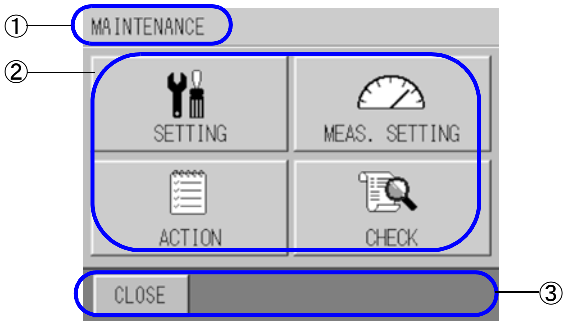


No.	Description	Display example
①	Title of display	MEAS., SETTING, LOG DATA, etc.
②	Clock	2004/01/30 09:57
③	Items to be measured	Abs, CONC, etc.
④	Readings	Instantaneous values
⑤	Conversion factor	Conversion factors for COD, etc.
⑥	Operation buttons	MAINTENANCE, CAL, etc.
⑦	Status of Unit	WARM UP! : Unit is being warmed up.
⑧	Range	Range
⑨	Switchable measurement items	Abs, CONC, etc.

WARM UP!

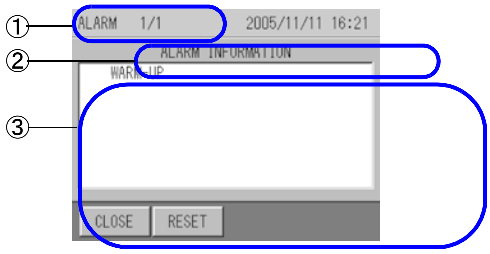
When the power is turned ON, “WARM UP!” is displayed. This will disappear after one hour.

Example of display - Displaying items to be selected



No.	Description	Display example
①	Title of display	MAINTENANCE, SETTING, etc.
②	Selection keys	SETTING, MEAS.SETTING, etc.
③	Switching keys	CLOSE, etc.

Example of display - Displaying history and other information



No.	Description	Display example
①	Title of display	Alarm history, calibration report, etc.
②	Items to be displayed	Description of alarms, etc.
③	Data list	-

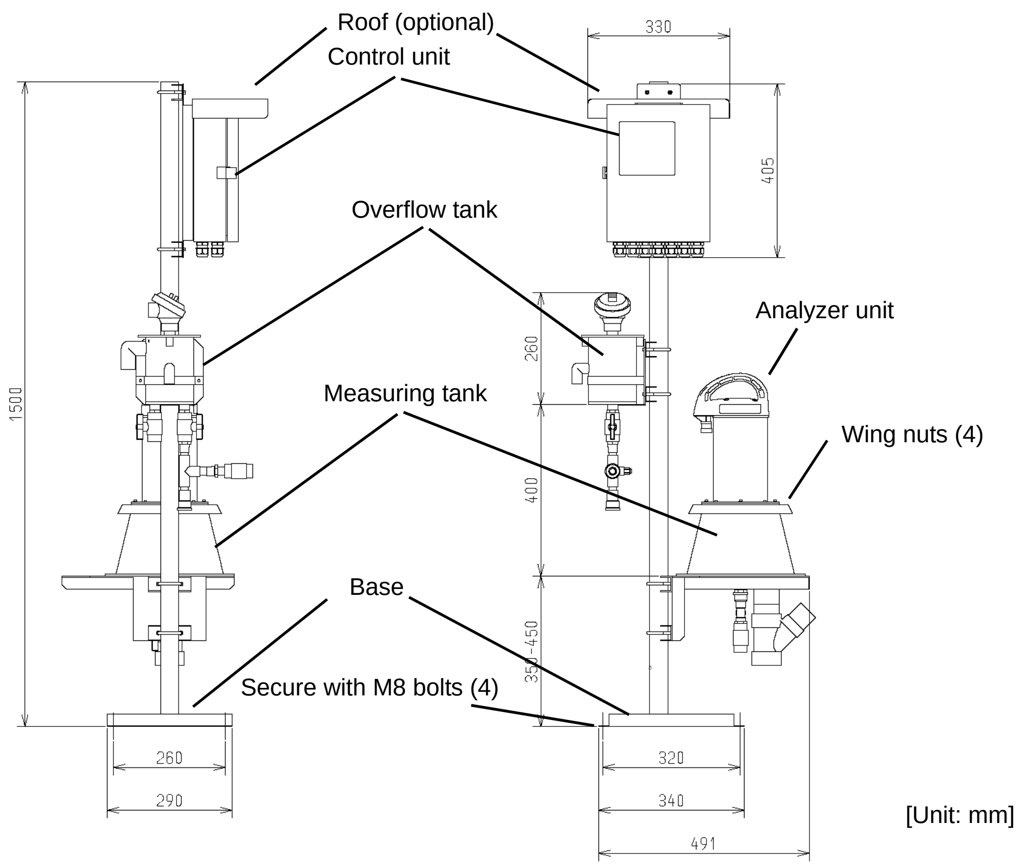
2 Installation

2.1 Conditions for Installation

- The unit is weather-proofed for outdoor installation; however, do not install the unit in places subject to direct sunlight.
- Install the unit near the water sampling point.
- Install the unit where a water-supply system, drainage facilities and a power supply are provided.
- Install the unit in a place that is well ventilated, free from dirt or dust, and where no corrosive gases will be generated.
- Install the unit in a place that is free from strong vibration or ferromagnetic fields.
- Install the unit where enough space is provided for conducting routine inspections and maintenance.
- Secure the unit using anchor bolts.
- Install the unit where enough space is provided for conducting services.

2.2 Installation Method

- Assemble the base, measuring tank, overflow tank and control unit as guided in the working instruction separately supplied.
- Secure the base with the reference bolts.



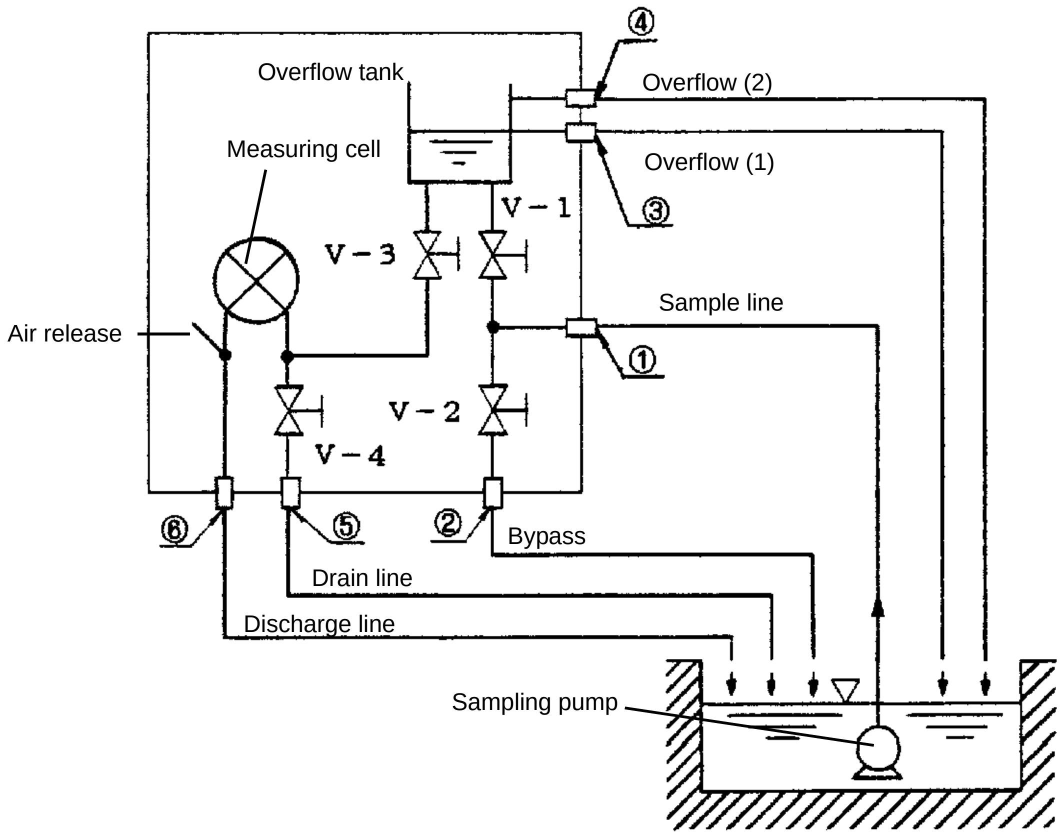
Note

The pole stand and the base illustrated are typical examples. Appearance and size may vary depending on specifications.

2.3 Piping Configuration

- Insulate the unit to protect against frozen pipes in the cold weather.
- Drain piping should be as short as possible in order not to create the backpressure. Make the drain cock discharge the air in order to prevent water from being blocked (rising pipes cannot be used).
- Diameters of piping sockets

	Piping description	Diameter of piping sockets
①	Sample inlet	Rp-1/2 female
②	Bypass outlet	Rc-1/2 female
③	Overflow outlet (1)	Elbow fitting, 13 dia. (nominal)
④	Overflow outlet (2)	Elbow fitting, 20 dia. (nominal)
⑤	Drain outlet	Rc-1/2 female
⑥	Discharge	Tube coupling, 50 dia. (nominal)



- Refer to the table below for the diameter for piping

	Piping description	Diameter of piping (when hard PVC piping is used)
①	Sample lines	Nominal diameter is 13 or larger
②	Bypass line	Nominal diameter is 13 or larger
③	Overflow line (1)	Nominal diameter is 13 or larger
④	Overflow line (2)	Nominal diameter is 20 or larger
⑤	Drain line	Nominal diameter is 13 or larger
⑥	Discharge line	Nominal diameter is 50 or larger

Note

- **Each piping diameter above is determined to go with the respective piping sockets of the OPSA-150. If the measuring unit is installed in an area far from the water sampling point and drainage point, each piping diameter should be larger than that indicated above.**
- **Ensure that the waste water piping, overflow piping (1) and (2), and drain piping have sufficient gradient to allow drainage from the outlet of each pipe. Ensure that the end of each piping is not immersed in water.**
- **When using soft PVC piping, use the braid reinforced flexible PVC (pressure-resistant) type.**
- **It is useful to use removable pipe tubing for regular cleaning.**

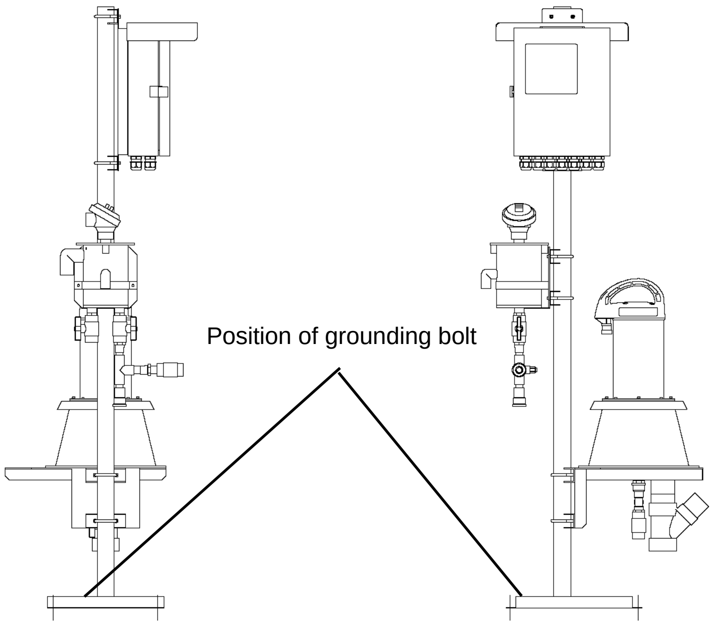
2.4 Wiring

⚠ WARNING

Maintain ground to avoid electric shock.

2.4.1 Grounding

- Be sure to ground the product with the ground resistance of 100 Ω or less.
- Make sure to ground to the grounding bolt.



Note

The pole stand and the base illustrated are typical examples Appearance and size may vary depending on specifications.

2.4.2 Power Supply

- Connect the power cord to the power source according to the signal table.
- Use a power cable of size 1.25 mm^2 (AWG16) or larger.
- Connect a surge absorber or noise killer in parallel to the power line in order to prevent electric power noise.

2.4.3 Connecting Signal Lines

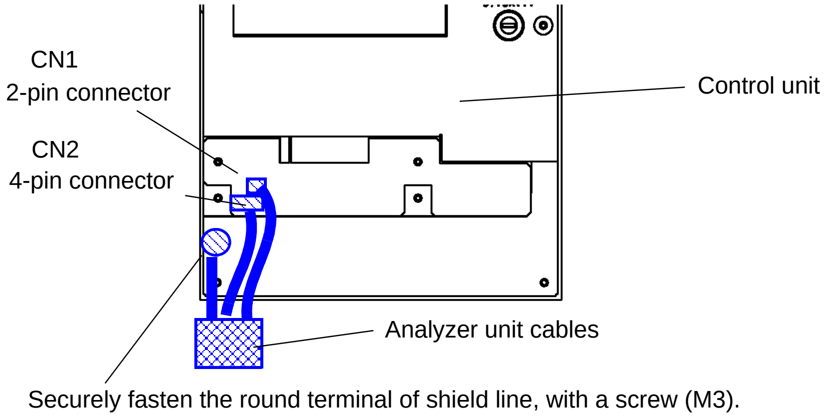
- Connect the signal lines according to the signal table.
- Use double shield cables for current output signal lines. The shield should be grounded at the receiver.
- Connect a surge absorber or noise killer in parallel to the contact output signal line in order to prevent electric power noise.

Note

Do not connect the power line to the signal line wiring. It may cause the unit malfunction.

2.4.4 Connecting Analyzer Unit to Control Unit

- Run the cable connectors of analyzer unit through the cable clamps at the lower left of control unit case.
- Connect the connectors of the analyzer unit to CN1 and CN2 at the lower left of the control unit inside.
- Securely fasten the round terminal of shield line, with a screw (M3), to the screw hole under the connectors CN1 and CN2.



2.4.5 Connecting Float Switch Cable of Overflow Tank to Control Unit

- Run the float switch cables of the overflow tank through the cable clamps at the lower left of control unit case.
- Connect the cables to the terminals No.7 and 8 at the lower center of control unit inside.

3 Operation

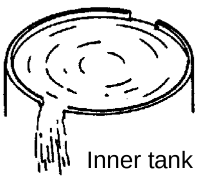
3.1 Preparation for Operation

1. **Ensure that the unit is installed properly according to “ 2 Installation ” (page 6).**
2. **Remove the lid of the overflow tank.**
3. **Remove the analyzer unit if the unit is already set to the measuring tank.**

Note

When removing from the measuring tank, put the analyzer unit on a flat surface to prevent it from falling. It is recommendable to put the analyzer unit on the calibration tank for safe and easy handling.

4. **Operate the valves as instructed in the flow sheet and dimensional outline drawing:**
   V-1 and V-4: close all (CLOSE)
   V-2 and V-3: open all (OPEN)
5. **Supply 2 L/min to 20 L/min of sample water from the underwater pump or the header.**
6. **Open valve (V-1) gradually until sample water runs out of the overflow outlet (1) of overflow tank. If the sample water does not run out of the overflow outlet (1) even when the valve (V-1) is fully opened, then close valve (V-2) gradually to adjust the flow from the overflow outlet (1).**
7. **Ensure that sample water overflows from the upper notches (2) of the inner tank of measuring tank and is discharged from the drain outlet.**



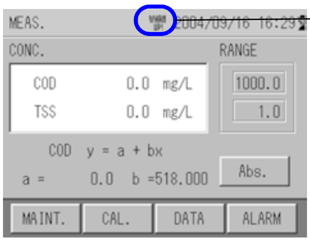
Inner tank of measuring tank

8. **When outlet flow from the overflow tank and the inner tank of measuring tank reaches the designated volume and keeps running consistently, attach the lid of the overflow tank and analyzer unit.**
9. **Make sure that the cables of analyzer unit and control unit are securely connected.**
10. **Set the switches of control panel as follows.**
    POWER switch: OFF
    Maintenance switch: ON

3.2 Starting Operation

1. **Turn the POWER switch ON.**

The MEAS. screen appears after a while.



Note

When the power is turned ON, the motor for analyzer unit wipers starts operation at low speed. Some noises and vibrations may be generated when the motor starts. Note that this is normal, and does not mean any malfunction of the unit.

"WARM UP!" continues to be indicated until the measured values become stable (for about 60 minutes).

⚠ WARNING

Do not touch the cell or directly view the UV lamp of the analyzer unit when you check the unit operation.

2. **Set the clock.**

Reference

" 5.4.5 Time Adjustment " (page 40)

3. **Change the settings as necessary.**

Reference

" 5 Functions " (page 32)

4. **Calibrate the unit.**

Note

Calibration should be performed after the measured value becomes stable. Before the calibration, perform the running-in operation for an hour or more using the sample water.

Reference

" 4 Calibration " (page 14)

Complete the procedure for the preparation of operation. The measurement starts.

5. **Turn OFF the maintenance switch.**

3.3 Shutting Down

3.3.1 For Shut Down within 7 Days

1. **Turn ON the maintenance switch.**
2. **Turn OFF the POWER switch.**

3.3.2 For Shut Down over 7 Days

1. **Close all the valves (V-1 to V-4).**
2. **Turn ON the maintenance switch.**
3. **Turn OFF the POWER switch.**
4. **Clean the overflow tank and measuring tank.**

Reference

“ 5.6 Maintenance - Action ” (page 52)

5. **Clean the cell of control unit.**

Reference

“ 2.3 Piping Configuration ” (page 7)

3.4 Restarting Operation

3.4.1 For Shut Down within 7 Days

Follow the procedure of “ 3.2 Starting Operation ” (page 12).

3.4.2 For Shut Down over 7 Days

1. **Follow the procedure of “ 3.1 Preparation for Operation ” (page 11).**
2. **Follow the procedure of “ 3.2 Starting Operation ” (page 12).**

4 Calibration

Calibration is the adjustment so that the value indicated by the unit corresponds to the actual value. It is necessary to obtain measurements of high accuracy and maintain the performance of the unit.

4.1 Calibration Pattern and its Cycle

It is necessary to perform the calibration periodically.
Perform calibration by following the cycle shown below.

◎ : Recommended cycle, ○ : Calibration as necessary

Measuring item	Calibration pattern	Targeted calibration cycle				Procedure Reference
		Before operation	Every week	Every month	When problems occur	
UV/VIS	Common zero calibration	◎	◎	◎	◎	page 20
	Common span calibration	◎	-	◎	◎	page 23
	Individual (UV) span calibration	○	-	○	○	page 26
TSS	Span calibration	◎	-	◎	◎	page 29

Perform the calibration in the order of "Common zero calibration," "Common span calibration," ("Individual (UV) span calibration" as necessary), and "TSS Span calibration."

● Descriptions for each Calibration

Calibration	Description
Common zero calibration	Common zero calibration of UV/VIS using the zero calibration solution (distilled water). Put the zero calibration solution in the calibration tank. After the measurement value has stabilized, perform calibration by operating the zero button.
Common span calibration	Common span calibration of UV/VIS using the exclusive common span calibration solution. Put the span calibration solution in the calibration tank. After setting the calibration value of UV/VIS and the measurement has stabilized, perform calibration by operating the span button. If performing either one of calibrations, set the span calibration value to "0" for the component that is not being calibrated. By doing this, the span calibration for one component becomes possible.
Individual (UV) span calibration	Individual span calibration of UV using the acid potassium phthalate standard solution (standard solution for UV calibration). Put the individual span calibration solution in the calibration tank. After setting the calibration value of UV and the measurement has stabilized, perform calibration by operating the span button. Make sure to set the span calibration value of VIS to "0" before performing the calibration.
Span calibration	Span calibration of TSS using the span calibration solution. Zero calibration is not required as the common zero calibration for UV/VIS is performed. Put the span calibration solution for TSS in the calibration tank. After setting the calibration value of TSS and the measurement has stabilized, perform calibration by operating the span button.

● About the Calibration coefficient

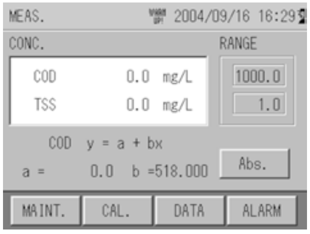
Calibration coefficient	Description
Zero calibration coefficient	Indicates misalignment of zero point
Span calibration coefficient	Indicates the gradient of sensitivity

4.2 Notes Regarding Calibration

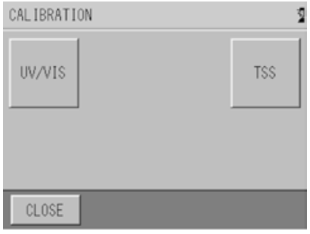
- Perform the calibration in the order of “Common zero calibration," "Common span calibration,” and “TSS Span calibration.”
- If the calibration coefficient exceeds the specified range during the calibration, an error occurs and the calibration value is not updated. The previous calibration value remains.
- Use distilled water for the common zero calibration solution. If a solution other than distilled water is used, the measurement error becomes larger in the low concentration range (0.1 Abs to 0.5 Abs).
  If the ion-exchanged water is used, the measurement error also becomes larger.
- Whenever diluting the span calibration solution, make sure to dilute by adding distilled water.
- When disposing of the common span calibration solution, follow the rules and regulations in your country/area. As the solution is colored, we recommend disposing of it by diluting with tap water unless otherwise specified.
- Store the common span calibration solution ampoule in a dark and cold place.
- Use the common span calibration solution ampoule immediately after opening.
- Do not ingest the common span calibration solution ampoule or its contents.
- If the common span calibration solution comes into contact with clothes, the clothes may be stained.
- Do not reuse the calibration solution. If the solution is altered, an accurate calibration may not be obtained.
- Prepare the span calibration solution by following the description in this manual.

4.3 Calibration Screen Display

1. **Press [CAL] on the MEAS. screen (main screen).**
   The CALIBRATION screen that shows the calibration menu is displayed.

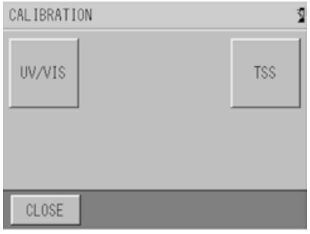


[CAL.]

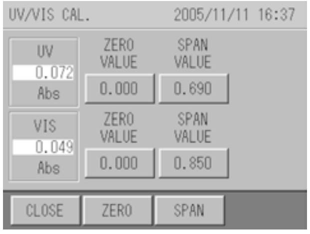


2. **Press the measurement item to be calibrated on the CALIBRATION screen.**
   The CALIBRATION screen for each measurement item is displayed.

Example when the calibration item is UV/VIS:



Calibration item:
[UV/VIS]



3. **Perform calibration by following the calibration procedure for each measurement item.**

4.4 Calibration of UV/VIS

There are three kinds of calibration method for UV/VIS: common zero calibration, common span calibration, and individual span calibration.

Reference

" 4.1 Calibration Pattern and its Cycle " (page 14).

4.4.1 How to Prepare Calibration Solution

Use distilled water for the zero calibration solution, and exclusive calibration solution for the common span calibration solution.

The calibration solution ampoule (H calibration solution (blue label)) for 1 Abs (100 m^{-1}) is attached to this unit. An attached calibration solution ampoule covers the measurement range up to 1 Abs. Full scale can be covered from 0.1 Abs to 5.0 Abs (10 m^{-1} to 500 m^{-1}) by using the 2 or 4 calibration solution ampoules.

Select the measurement full scale you use from the following, and compound the calibration solution.

Calibration solution	Measurement full scale range	Qty. of calibration solution ampoule	Concentration of calibration solution when diluted with 2 L
H calibration solution(blue label) 0 to 1 Abs for calibration	0.1 Abs to 1.0 Abs (10 m^{-1} to 100 m^{-1})	1	UV: 0.690 Abs (69.0 m^{-1}) VIS: 0.850 Abs (85.0 m^{-1})
	1.1 Abs to 2.5 Abs (110 m^{-1} to 250 m^{-1})	2	UV: 1.380 Abs (138.0 m^{-1}) VIS: 1.700 Abs (170.0 m^{-1})
	2.6 Abs to 5.0 Abs (260 m^{-1} to 500 m^{-1})	4	UV: 2.760 Abs (276.0 m^{-1}) VIS: 3.400 Abs (340.0 m^{-1})

Also, our calibration solution ampoule (H II calibration solution (green label)) for 2 Abs (200 m^{-1}) is also available. If using in the measurement full scale of 1.1 Abs (110 m^{-1}) or over, it is more economical to buy H II calibration solution ampoules as the number of ampoules used is fewer.

Calibration solution	Measurement full scale range	Qty. of calibration solution ampoule	Concentration of calibration solution when diluted with 2 L
H II calibration solution (green label) 0 to 2 Abs for calibration	1.1 Abs to 2.5 Abs (110 m^{-1} to 250 m^{-1})	1	UV: 1.380 Abs (138.0 m^{-1}) VIS: 1.700 Abs (170.0 m^{-1})
	2.6 Abs to 5.0 Abs (260 m^{-1} to 500 m^{-1})	2	UV: 2.760 Abs (276.0 m^{-1}) VIS: 3.400 Abs (340.0 m^{-1})

- **Common zero calibration solution**

Preparation

Distilled water: 2 L

Make sure to use the distilled water.

- **How to prepare common span calibration solution**

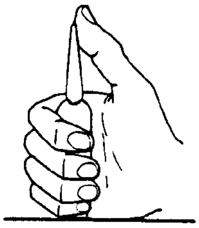
Preparation

Calibration solution ampoule:	1 to 4
Zero solution (distilled water):	approximately 3 L
2 L Measuring flask:	1
Cleaning bottle:	1

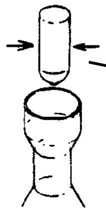
1. **Drain the solution remaining in the top of ampoule to the bottom part.**

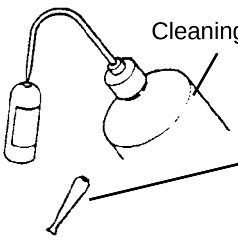
2. **Make a cut line on the ampoule with a cutter knife, knife or other sharp object.**

3. **Break the ampoule.**



4. **Pour the contents from the ampoule into a 2 L measuring flask that was cleaned using zero solution in advance.**





5. **Pour the zero solution up to the gauge line of 2 L measuring flask, and dissolve the solution by shaking thoroughly.**

Note

- **Make sure to use distilled water to dilute the span calibration solution. Do not use the tap water, groundwater, or industrial water.**
- **Use 2 ampoules of H calibration solution when preparing the span calibration solution of 1.1 Abs to 2.5 Abs (110 m^{-1} to 250 m^{-1}). In this case, the concentration value of span calibration solution is double the concentration value described on the ampoule.**
- **Use 4 ampoules of H calibration solution or 2 ampoules of H II calibration solution when preparing the span calibration solution of 2.6 Abs to 5.0 Abs (260 m^{-1} to 500 m^{-1}). In this case, the concentration value of span calibration solution is 4 times the concentration value described on the ampoule for H calibration solution, and double the H II calibration solution ampoule.**

● How to prepare span calibration solution of acid potassium phthalate standard solution (standard solution for UV calibration)

Normally calibration with the common span calibration solution is available; also, UV calibration with acid potassium phthalate is possible.

Preparation

Acid potassium phthalate:	using the special grade chemicals, dried for 1 hour at a temperature of 120°C, and cooled in desiccators
Zero solution (distilled water):	approximately 3 L
1 L Beaker:	1
2 L Measuring flask:	1
Electronic balance:	1 (Accuracy shall be within 1 mg.)
Cleaning bottle:	1

1. **Determine the requirement of acid potassium phthalate from the following table according to the measurement range.**

UV measurement range	Requirement of acid potassium phthalate when compounded in 2 L	Concentration of span calibration solution	Light absorption when the temperature of calibration solution is 25°C (absorption factor)	Light absorption when the temperature of calibration solution is 20°C (absorption factor)	Light absorption when the temperature of calibration solution is 30°C (absorption factor)
0.1 Abs to 0.5 Abs ($10\ m^{-1}$ to $50\ m^{-1}$)	100 mg	50 mg/L	0.44 Abs ($44\ m^{-1}$)	0.43 Abs ($43\ m^{-1}$)	0.45 Abs ($45\ m^{-1}$)
0.6 Abs to 1.0 Abs ($60\ m^{-1}$ to $100\ m^{-1}$)	200 mg	100 mg/L	0.87 Abs ($87\ m^{-1}$)	0.85 Abs ($85\ m^{-1}$)	0.89 Abs ($89\ m^{-1}$)
1.1 Abs to 2.5 Abs ($110\ m^{-1}$ to $250\ m^{-1}$)	400 mg	200 mg/L	1.74 Abs ($174\ m^{-1}$)	1.70 Abs ($170\ m^{-1}$)	1.78 Abs ($178\ m^{-1}$)
2.6 Abs to 5.0 Abs ($260\ m^{-1}$ to $500\ m^{-1}$)	800 mg	400 mg/L	3.48 Abs ($348\ m^{-1}$)	3.40 Abs ($340\ m^{-1}$)	3.56 Abs ($356\ m^{-1}$)

Note

The calibration solution of acid potassium phthalate has a temperature characteristic. The temperature characteristic is (4.5×10^{-3})/°C at a temperature of 5°C to 30°C. Measure the temperature of calibration solution to calculate the light absorption during calibration.
Calculation method:
Light absorption of calibration solution = Light absorption of calibration solution at a temperature of 25°C × (1 + (temperature of calibration solution − 25) × 0.0045)
(The above table shows light absorptions also at 20°C and 30°C as a reference.)

2. **Measure the requirement of acid potassium phthalate (special grade chemicals) following the table above.**

3. **Pour approximately 500 mL of the zero solution (distilled water) into the 1 L beaker, and add the measured acid potassium phthalate to dissolve thoroughly.**

Tip

The acid potassium phthalate dissolves easily when setting the temperature of the zero solution from 40°C to 50°C.

4. **Pour the solution in the 1 L beaker into the 2 L measuring flask, add zero solution to the gauge line, and shake to dissolve the solution.**

4.4.2 Common Zero Calibration

[CAL.] [UV/VIS]

MEAS. screen → CALIBRATION screen → UV/VIS CAL. screen

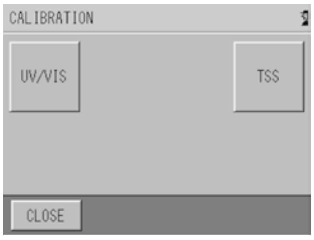
Refer to " 4.3 Calibration Screen Display " (page 16)

● Preparation

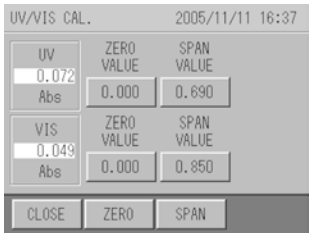
Plastic bucket:	1 (diameter: approximately 30 cm)
Zero solution (distilled water):	approximately 3 L
Calibration tank:	1
2 L Measuring flask:	1
Cleaning bottle:	1

● Calibration procedure

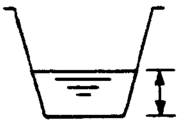
1. **Press [UV/VIS] on the CALIBRATION screen of control unit.**



Calibration item [UV/VIS]



2. **Clean the attached calibration tank with zero solution thoroughly, and pour the zero solution to the scale.**
3. **Prepare the plastic bucket and pour the tap water.**



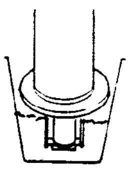
Pour the tap water to a depth of approximately 9 cm
(so that the base of the analyzer unit may not soak in the water).

4. **Remove the analyzer unit from the measurement tank.**

Note

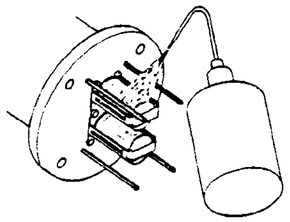
- **When removing the analyzer unit, make sure not to lose the wing nuts.**
- **As the UV lamp is illuminated, work with an eye protector. The UV lamp should not be viewed directly as your eyes may be damaged.**

5. **Clean the cell unit of the analyzer unit.**
   - If there is any extraneous matter around the cell, remove it.
   - Next, dip the cell unit in the plastic bucket to remove contamination.



Dip the cell unit in the water and shake it from side to side
and up and down to remove contamination.

- Clean the cell unit using the zero solution in the cleaning bottle again.



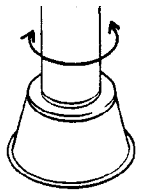
Clean the cell unit using the zero solution in the cleaning bottle.

- If the cell unit is contaminated heavily, clean the cell unit by referring to “ 7.2.1 Cleaning Method for Measuring Cell ” (page 80).

Note

Do not spill any solution on the base of the cell rotary shaft (V-ring seal part).

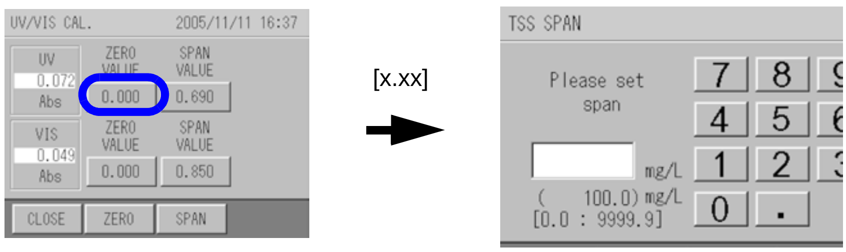
6. After cleaning the cell unit, dip the cell in the calibration tank.



Allow the solution to adhere by shaking from side to side for several times.

7. Press the button for [ZERO VALUE] for UV on the UV/VIS CAL. screen.

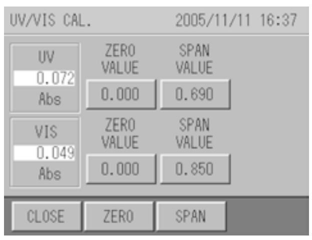
The screen changes to the UV ZERO screen.



8. Enter "0" with numeric keypad, and press [SET].

Input range: 0.000 Abs to 5.000 Abs or 0.0 m^{-1} to 500.0 m^{-1}

The screen returns to the UV/VIS CAL. screen.



Note

The decimal place of input range varies according to the setting in “ 5.5.2 Setting of Decimal Place for each Component ” (page 45).

Tip

[CLEAR]: Clears all input numeric values.
[BACK]: Clears the rightmost number.

9. **Press the button for [ZERO VALUE] for VIS on the UV/VIS CAL. screen.**

10. **Enter "0" with numeric keypad, and press [SET].**

Input range: 0.000 Abs to 5.000 Abs or 0.0 m^{-1} to 500.0 m^{-1}

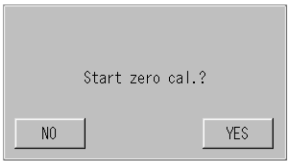
The screen returns to the UV/VIS CAL. screen.

Note

Though any number can be entered for the zero calibration value, normally, enter "0".

11. **After waiting for 3 minutes and confirming that the measurement value has stabilized, press [ZERO].**

The “Start zero cal.?” screen is displayed.



12. **Press [YES] to start the zero calibration.**

The “Calibrating” screen is displayed.

After calibration, the “Calibration finished” screen is displayed.

13. **Press [YES].**

The zero calibration value for UV/VIS is updated.

14. **The common zero calibration is complete.**

If performing the common span calibration subsequently, start the common span calibration with the analyzer unit still in the zero calibration tank.

If finishing calibration, set the analyzer unit in the measuring tank.

Note

- **When the measurement value increases gradually after stabilization, the solution may be contaminated. Replace the zero solution and calibrate again.**
- **If the temperature difference of the sample water and zero solution is large, the measurement value may be unstable. Calibrate under the conditions where the temperature difference is as small as possible.**

4.4.3 Common Span Calibration

[CAL.] [UV/VIS]

MEAS. screen → CALIBRATION screen → UV/VIS CAL. screen

Refer to “ 4.3 Calibration Screen Display ” (page 16)

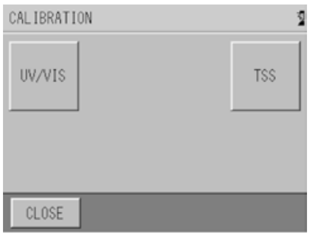
Perform the common span calibration after finishing the common zero calibration.

- **Preparation**

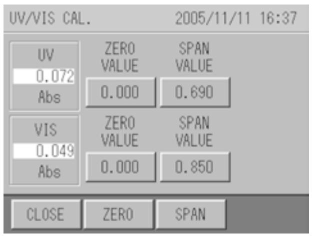
Calibration tank:	1
Span calibration solution:	2 L (measuring flask)
Cleaning bottle:	1

- **Calibration procedure**

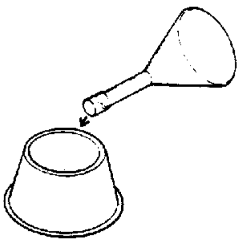
1. **Press [UV/VIS] on the CALIBRATION screen of the control unit.**



Calibration item
[UV/VIS]



2. **Clean the attached calibration tank with zero solution thoroughly, pour approximately 100 mL of span solution (in the 2 L measuring flask), which is prepared by following “ 4.4.1 How to Prepare Calibration Solution ” (page 17), and clean the tank inside.**



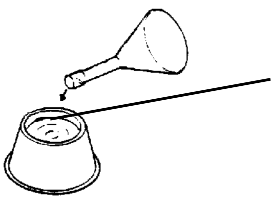
Pour approximately 100 mL
of span solution.

Shake up and down.

Clean the tank inside.

Discard the solution.

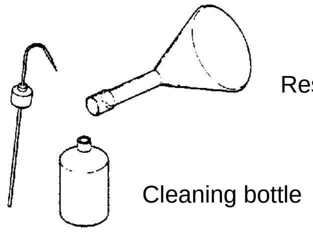
3. **After cleaning, pour the span solution up to the scale.**



Scale

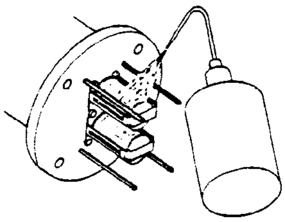
Amount to the scale is approximately 1.3 L.

4. **Pour the residual common span solution into the cleaning bottle.**



When performing span calibration, the calibration work can be performed smoothly if the preparation through step 4. is performed together with the preparation for the common zero calibration.

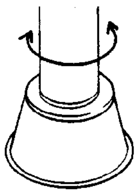
5. **Take the analyzer unit from the zero calibration tank, and clean the cell unit using the common span solution in the cleaning bottle .**



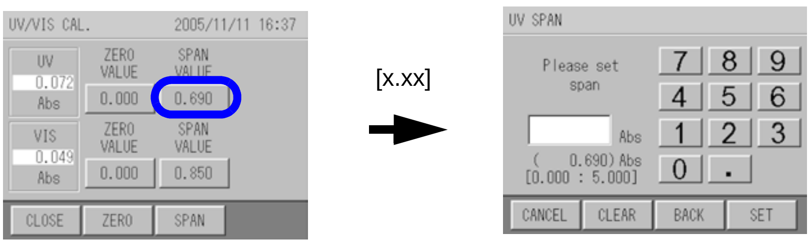
Note

Do not spill any solution on the base of cell rotary shaft (V-ring seal part).

6. **After cleaning the cell unit with the common span solution, dip it in the calibration tank.**



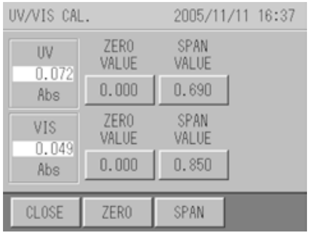
7. **Press the button for [SPAN VALUE] for UV on the UV/VIS CAL. screen.**
   The screen changes to the UV SPAN screen.



8. Enter the concentration value of compounded span calibration solution with the numeric keypad, and press [SET].

Input range: 0.000 Abs to 5.000 Abs or 0.0 m^{-1} to 500.0 m^{-1}

The screen returns to the UV/VIS CAL. screen.



Note

- **The decimal place of the input range varies according to the setting in " 5.5.2 Setting of Decimal Place for each Component " (page 45).**
- **Refer to " 4.4.1 How to Prepare Calibration Solution " (page 17) for the concentration value of compounded span calibration solution.**
- **If performing either one of the calibrations, set the span calibration value to "0" for the component that is not being calibrated. By doing this, the span calibration for single component becomes possible.**

Tip

[CLEAR]: Clears all input numeric values.
[BACK]: Clears the rightmost number.

9. Press the button for [SPAN VALUE] for VIS on the UV/VIS CAL. screen.

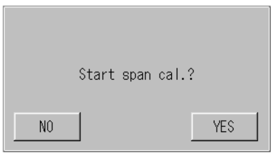
10. Enter the concentration value of compounded span calibration solution with the numeric keypad, and press [SET],

Input range: 0.000 Abs to 5.000 Abs or 0.0 m^{-1} to 500.0 m^{-1}

The screen returns to the UV/VIS CAL. screen.

11. After waiting for 3 minutes and confirming that the measurement value is stabilized, press [SPAN].

The "Start span cal.?" screen is displayed.



12. Press [YES] to start the span calibration.

The "Calibrating" screen is displayed.

After calibration, the "Calibration finished" screen is displayed.

13. Press [YES].

The span calibration value for UV/VIS is updated.

14. The common span calibration is finished.

If performing another span calibration subsequently, start the span calibration with the analyzer unit still in the calibration tank.

If finishing the calibration, set the analyzer unit in the measuring tank.

Note

Do not reuse the calibration solution used in the calibration tank. The accuracy of calibrated value may be lost.

4.4.4 Individual (UV) Span Calibration

[CAL.] [UV/VIS]

MEAS. screen → CALIBRATION screen → UV/VIS CAL. screen

Refer to “ 4.3 Calibration Screen Display ” (page 16)

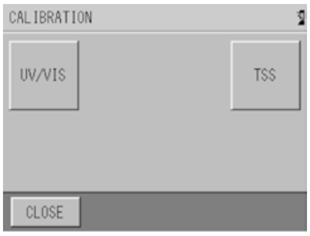
Perform the individual (UV) span calibration after finishing the common span calibration.

- **Preparation**

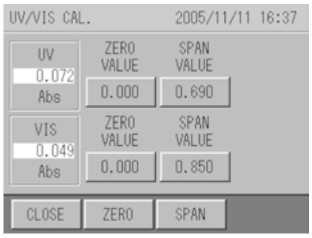
Calibration tank:	1
Span calibration solution:	2 L (measuring flask)
Cleaning bottle:	1

- **Calibration procedure**

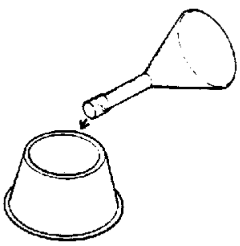
1. **Press [UV/VIS] from the CALIBRATION screen of control unit.**



Calibration item
[UV/VIS]

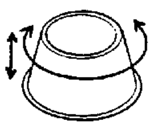


2. **Clean the attached calibration tank with zero solution thoroughly, pour approximately 100 mL of the span calibration solution of acid potassium phthalate (from the 2 L measuring flask), which is prepared by following “ 4.4.1 How to Prepare Calibration Solution ” (page 17), and clean the tank inside.**



Pour approximately 100 mL
of span solution.

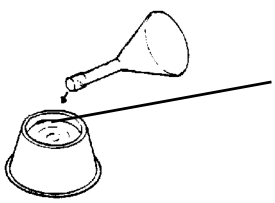
Shake up and down.



Clean the tank inside.

Discard the solution.

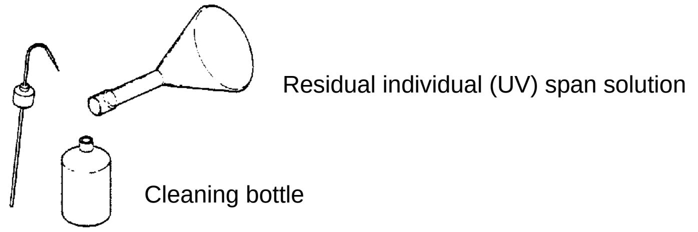
3. **After cleaning, pour the individual span solution to the scale.**



Scale

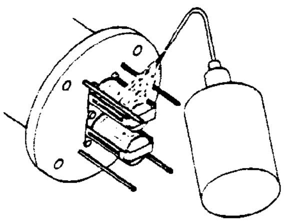
Amount to the scale is approximately 1.3 L.

4. **Pour the residual individual span solution into the cleaning bottle.**



When performing span calibration, the calibration work can be performed smoothly if the preparation through step 4. is performed together with the preparation for the common zero calibration.

5. **Take the analyzer unit from the zero calibration tank, and clean the cell unit using the cleaning bottle containing the individual span solution.**

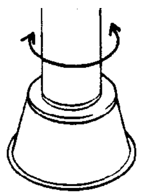


Clean the cell unit using the cleaning bottle containing the individual (UV) span solution.

Note

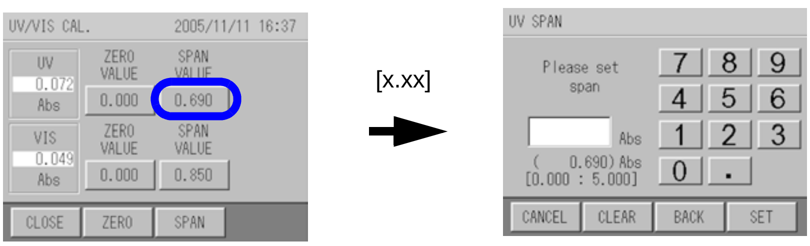
Do not spill any solution on the base of the cell rotary shaft (V-ring seal part).

6. **After cleaning the cell unit with the individual span solution, dip in the calibration tank.**

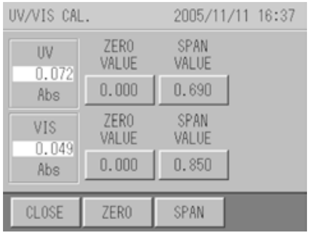


Allow the solution to adhere by shaking from side to side for several times.

7. **Press the button for [SPAN VALUE] for UV on the UV/VIS CAL. screen.**
The screen changes to the UV SPAN screen.



8. **Enter the concentration value of compounded span calibration solution with the numeric keypad, and press [SET].**
Input range: 0.000 Abs to 5.000 Abs or 0.0 m^{-1} to 500.0 m^{-1}
The screen returns to the UV/VIS CAL. screen.



Note

- **The decimal place input range varies according to the setting in "5.5.2 Setting of Decimal place for each Component" on page 45.**
- **Refer to “ 4.4.1 How to Prepare Calibration Solution ” (page 17) for the concentration value of compounded span calibration solution.**

Tip

[CLEAR]: Clears all input numeric values.
[BACK]: Clears the rightmost number.

9. **Press the button for [SPAN VALUE] for VIS on the UV/VIS CAL. screen.**

10. **Enter "0" with the numeric keypad, and press [SET],**
Input range: 0.000 Abs to 5.000 Abs or 0.0 m^{-1} to 500.0 m^{-1}
The screen returns to the UV/VIS CAL. screen.

11. **After waiting for 3 minutes and confirming that the measurement value has stabilized, press [SPAN].**
The “Start span cal.?” screen is displayed.

12. **Press [YES] to start the span calibration.**
The “Calibrating” screen is displayed.
After calibration, the “Calibration finished” screen is displayed.

13. **Press [YES].**
The span calibration value for UV/VIS is updated. VIS calibration value is not updated.

14. **The individual span calibration is finished.**
If performing TSS span calibration subsequently, start the TSS span calibration with the analyzer unit still in the calibration tank.
If finishing the calibration, set the analyzer unit in the measuring tank.

Note

Do not reuse the calibration solution used in the calibration tank. The accuracy of calibrated value may be lost.

4.5 TSS Span Calibration

Only the span calibration is applied to the calibration method for TSS.
Perform TSS span calibration after performing the common zero calibration and the common span calibration of UV/VIS.
Refer to “ 4.1 Calibration Pattern and its Cycle ” (page 14).
TSS span calibration is to obtain the TSS coefficient converted from the light absorption of VIS.
Use measurement sample for span solution.
The rough standard of the TSS measuring range is approximately 200 mg/L.

4.5.1 TSS Span Calibration

[CAL.] [UV/VIS]

MEAS. screen → CALIBRATION screen → TSS CAL screen

Refer to “ 4.3 Calibration Screen Display ” (page 16)

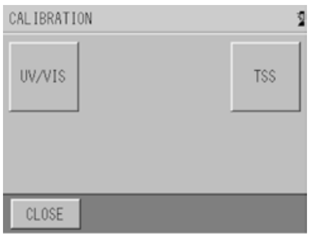
Perform the TSS span calibration after finishing the common zero calibration and the common span calibration.

● **Preparation**

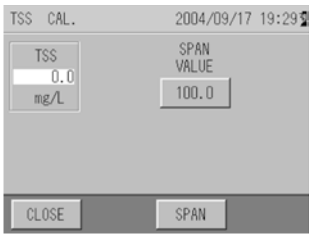
Calibration tank:	1
Span calibration solution:	2 L (measuring flask)
Cleaning bottle:	1

● **Calibration procedure**

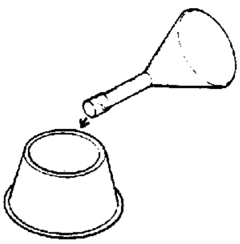
1. **Press [TSS] on the CALIBRATION screen of the control unit.**



Calibration item
[TSS]



2. **Clean the attached calibration tank with zero solution thoroughly, pour approximately 100 mL of the TSS span solution (from the 2 L measuring flask).**



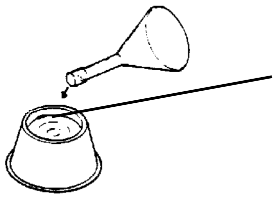
Pour approximately 100 mL
of span solution.

Shake up and down.

Clean the tank inside.

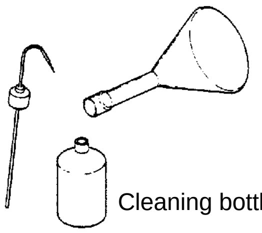
Discard the solution.

3. **After cleaning, pour the TSS span solution to the scale.**



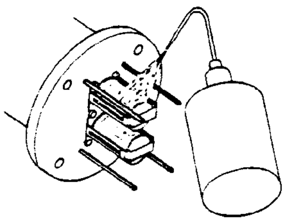
Scale
Amount to the scale is approximately 1.3 L.

4. **Pour the residual TSS span solution into the cleaning bottle.**



Residual TSS span solution
Cleaning bottle

5. **Take the analyzer unit from the calibration tank, and clean the cell unit using the TSS span solution in the cleaning bottle.**

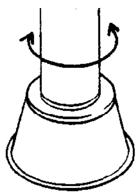


Clean the cell unit using the TSS span solution in the cleaning bottle.

Note

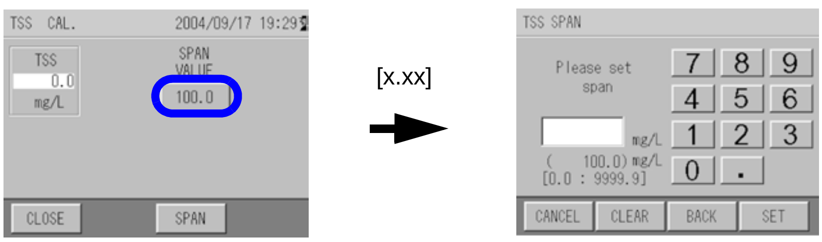
Do not spill any solution on the base of cell rotary shaft (V-ring seal part).

6. **After cleaning the cell unit with the TSS span solution, dip in the calibration tank.**



Allow the solution to adhere by shaking from side to side for several times.

7. **Press the button for [SPAN VALUE] on the TSS CAL. screen.**
The screen changes to the TSS SPAN screen.



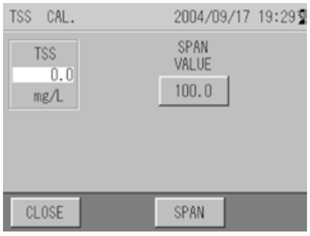
8. Enter the concentration value of TSS calibration solution with numeric keypad, and press [SET],

Input range: 0.00 to 9999.99

The screen returns to the TSS CAL. screen.

Note

The decimal place of input range varies according to the setting in “ 5.5.2 Setting of Decimal Place for each Component ” (page 45).



Note

Enter the concentration value of calibration solution for TSS calibration value.

Tip

[CLEAR]: Clears all input numeric value.
[BACK]: Clears the rightmost number.

9. After waiting for 3 minutes and confirming that the measurement value is stabilized, press [SPAN].

The “Start span cal.?” screen is displayed.

10. Press [YES] to start the span calibration.

“Calibrating” screen is displayed.

After calibration, “Calibration finished” screen is displayed.

11. Press [YES].

The span calibration value for TSS is updated.

12. The TSS span calibration is finished.

Set the analyzer unit in the measuring tank.

Note

Do not reuse the calibration solution used in the calibration tank. The accuracy of calibrated value may be lost.

5 Functions

This unit has various functions. It is necessary to set various conditions to utilize these functions fully. This chapter describes the functions and the setting procedures.

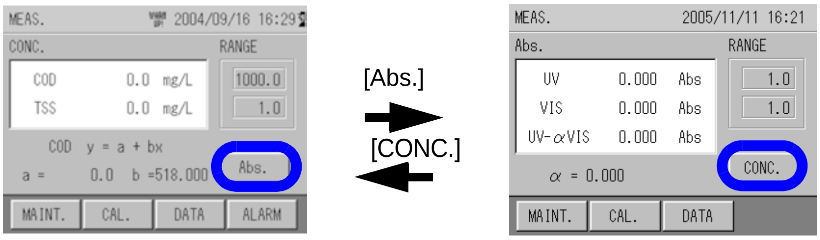
5.1 MEAS. (Measurement) Screen

The main screen displayed at the start when the power in ON.

Measurement screen contains two screens; the “CONC.” screen to display the COD concentration and TSS concentration, and the “Abs.” screen to display the light absorption of UV, VIS, and UV-αVIS.

To switch screens, press the [Abs.] and [CONC.] buttons.

When the power is turned ON the CONC. screen is displayed. The previous measurement item is memorized when the menu is switched.



5.2 Table of Functions

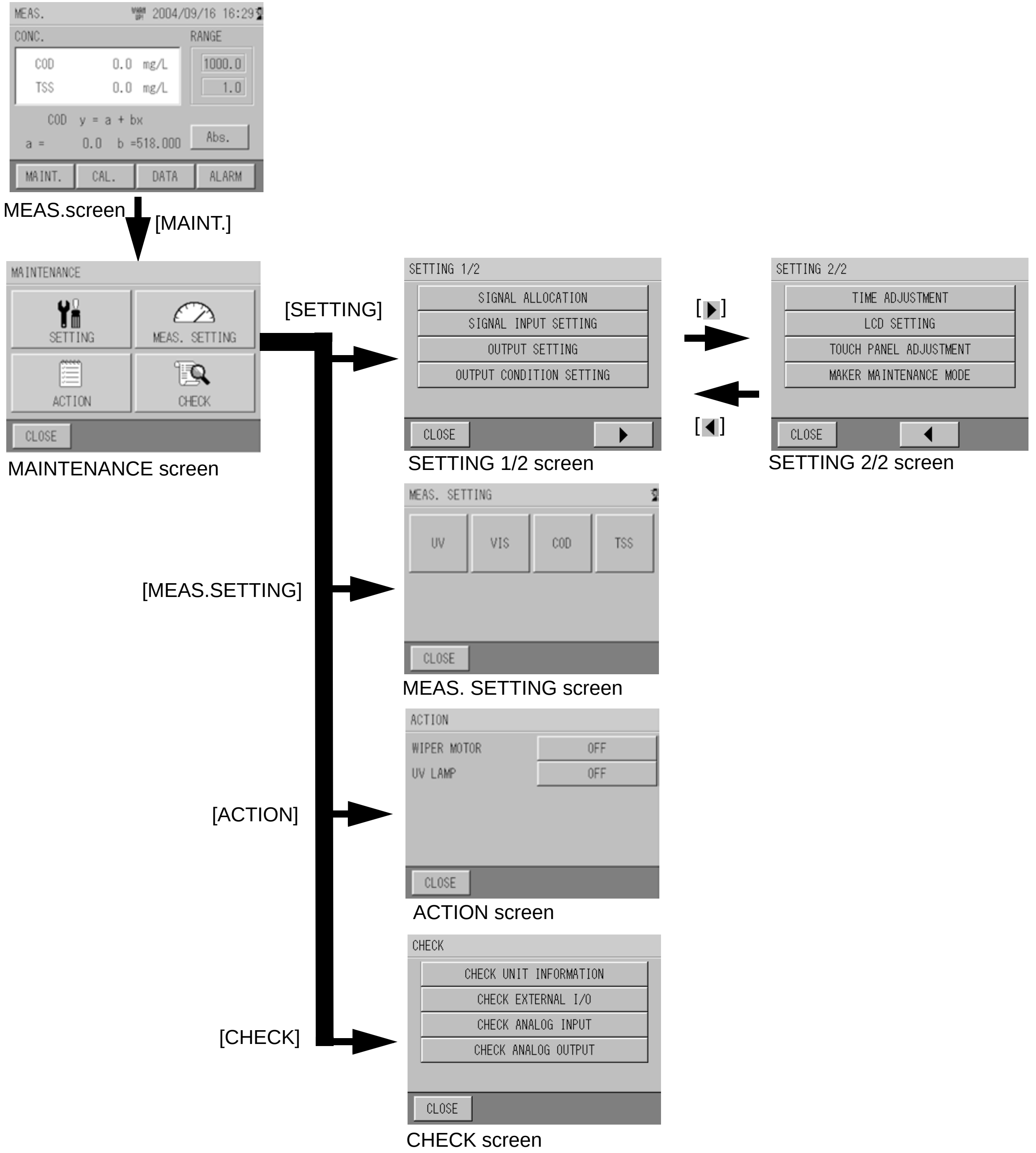
Thoroughly understand the functions and set the functions so as to meet the customer’s needs.

Function		Description	Reference
Maintenance	Setting	Sets signal reception and allocation from exterior, and input and output setting during operation.	“ 5.4 Maintenance - Setting ” (page 34)
	Measurement setting	Sets the measurement range, unit, and upper and lower limit for UV, VIS, COD, and TSS.	“ 5.5 Maintenance - Measurement Setting ” (page 42)
	Action	Checks ON/OFF of motor for wipers and UV lamp.	“ 5.6 Maintenance - Action ” (page 52)
	Check	Checks the unit information and input and output operation.	“ 5.7 Maintenance - Check ” (page 53)
Data	Log data	Checks the measurement data saved in the memory.	“ 5.8 Data Check ” (page 56)
	Graph	Displays graphs of the measurement data saved in the memory.	
	Calibration report	Displays the calibration report saved in the memory.	
	CF card	Saves the memorized data on the CF card or initializes the CF card	
Alarm	Alarm	Immediately displays the alarm being triggered.	“ 5.9 Alarm ” (page 65)
	Alarm history	Displays the alarm history saved in the memory.	

5.3 Maintenance Screen Display

1. **Press [MAINT.] on the MEAS. screen.**
   The MAINTENANCE screen appears.
2. **Select the maintenance item with buttons displayed on the MAINTENANCE screen.**
   The MAINTENANCE screen for each item appears.

MAINTENANCE screen operation buttons	Displayed screen
[SETTING]	SETTING screen
[MEAS. SETTING]	MEAS. SETTING screen
[ACTION]	ACTION screen
[CHECK]	CHECK screen



5.4 Maintenance - Setting

All maintenance settings are performed here.

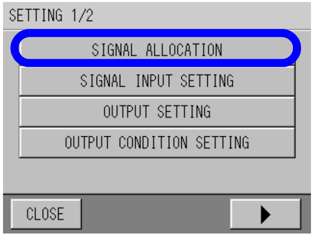
5.4.1 Signal Allocation

MEAS. screen → [MAINT.] → MAINTENANCE screen → [SETTING] → SETTING 1/2 screen → [SIGNAL INPUT SETTING] → SIGNAL ALLOCATION screen (1/3)

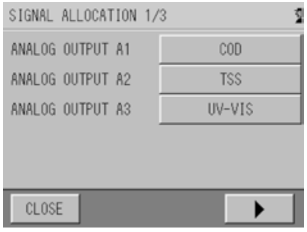
Refer to " 5.3 Maintenance Screen Display " (page 33).

The output terminals for analog output and contact output are allocated here.

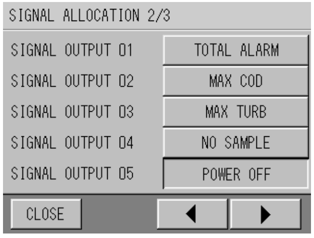
1. **Press [SIGNAL ALLOCATION] on SETTING 1/2 screen.**
The SIGNAL ALLOCATION screen appears.



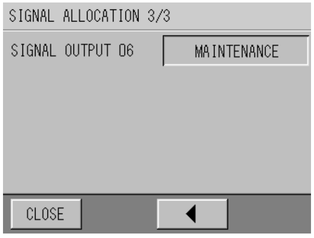
[SIGNAL ALLOCATION]



[▶]



[▶]



2. Output signal is allocated.

Setting item	Description	Initial setting
ANALOG OUTPUT A1	Sets to output either one of the following measurement component to the analog output terminal from A1 to A3. Setting range: UV, VIS, UV-VIS, COD, TSS, NONE	COD
ANALOG OUTPUT A2		TSS
ANALOG OUTPUT A3		UV-VIS
SIGNAL OUTPUT 01	Sets to output either one of the following items to the output terminal of contact output from 01 to 04. Setting range: TOTAL ALARM, MAX COD, MAX TSS, MIN COD, MIN TSS, LAMP ERR., NO SAMPLE, MOTOR ERR., ANALYZE ERR., WARM-UP	TOTAL ALARM
SIGNAL OUTPUT 02		MAX COD
SIGNAL OUTPUT 03		MAX TSS
SIGNAL OUTPUT 04		NO SAMPLE
SIGNAL OUTPUT 05	Outputs the following item to the output terminal 05 of contact output. POWER OFF (cannot be changed)	POWER OFF (fixed)
SIGNAL OUTPUT 06	Outputs the following item to the output terminal 06 of contact output. MAINTENANCE (cannot be changed)	MAINTENANCE (fixed)

Reference

“ 6 External Input and Output ” (page 68)

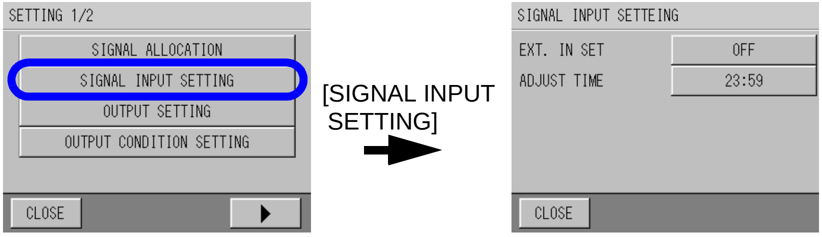
5.4.2 Signal Input Setting

	[MAINT.]		[SETTING]		[SIGNAL INPUT SETTING]	
MEAS. screen	→	MAINTENANCE screen	→	SETTING 1/2 screen	→	SIGNAL INPUT SETTING screen

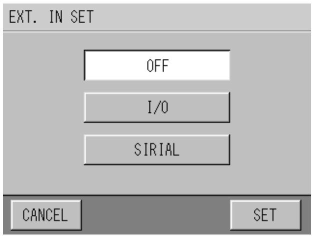
Refer to “ 5.3 Maintenance Screen Display ” (page 33).

Here enables clock time correction setting, such as setting external input of clock time and setting correct clock time.

1. **Press [SIGNAL INPUT SETTING] on the SETTING 1/2 screen.**
   The SIGNAL INPUT SETTING screen appears.



2. **Press the button for EXT. IN SET.**
   EXT. IN SET screen appears.

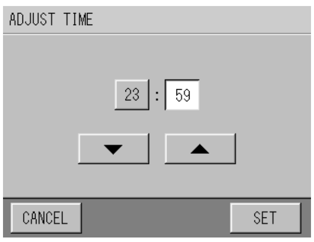


3. **Select the setting item and press [SET].**

Setting item	Description
OFF	Receives button operation only.
I/O	Receives the input via contacts.
SERIAL	Receives the input via RS-232C.

   The screen returns back to the SIGNAL INPUT SETTING screen.
   Setting for the system to receive the signal input is complete.

4. **Press the button for ADJUST TIME.**
   ADJUST TIME screen appears.



5. **Input the correct clock time when the external input is received.**
   The screen returns back to the SIGNAL INPUT SETTING screen.
   Setting the clock time (when the external input is received) is complete.

Note

Only the time within ±20 minutes of current set time can be input to set the correct clock time. If the time out of this range is input, the setting becomes invalid.

5.4.3 Output Setting

[MAINT.]	[SETTING]	[OUTPUT SETTING]	
MEAS. screen →	MAINTENANCE screen →	SETTING 1/2 screen →	OUTPUT SETTING screen

Refer to “ 5.3 Maintenance Screen Display ” (page 33).

Here enables the output parameter setting.

Setting item	Description	Initial setting
ANALOG OUTPUT LEVEL	Sets the analog output level. Setting range: 0 mA to 16 mA, 4 mA to 20 mA	4 mA to 20 mA
MAINT. OUTPUT	Sets the measurement output during calibration/maintenance to output by one of the settings below. Setting range: HOLD, DIRECT, PRESET (0 mA to 21 mA)	HOLD
WARM-UP OUTPUT	Sets the measurement output during warm-up to output by one of the settings below. Setting range: HOLD, DIRECT, PRESET (0 mA to 21 mA)	PRESET 4.0 mA
ALARM OUTPUT	Sets the measurement output during an alarm to output by one of the settings below. Setting range: HOLD, DIRECT, PRESET (0 mA to 21 mA)	HOLD

1. **Press [OUTPUT SETTING] on the SETTING 1/2 screen.**
   The OUTPUT SETTING screen appears.
2. **Press the the button for ANALOG OUTPUT LEVEL.**
   The ANALOG OUTPUT LEVEL screen appears.
3. **Select the setting item from [0 - 16] or [4 - 20] and press [SET].**
   The screen returns back to the OUTPUT SETTING screen.
   Setting for the analog output level is complete.

● Output settings during operation

Here describes the following three output setting methods.

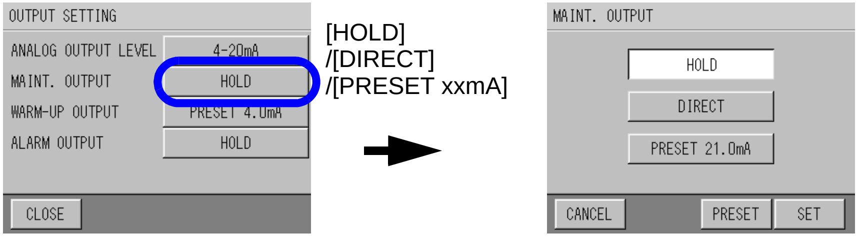
- MAINT. OUTPUT
- WARM-UP OUTPUT
- ALARM OUTPUT

[MAINT.]	[SETTING]	[OUTPUT SETTING]	
MEAS. screen →	MAINTENANCE screen →	SETTING 1/2 screen →	OUTPUT SETTING screen

Refer to “ 5.3 Maintenance Screen Display ” (page 33).

1. **Press the output item button to be set on the OUTPUT SETTING screen.**
   - MAINT. OUTPUT
   - WARM-UP OUTPUT
   - ALARM OUTPUT

   The screens for the output of each item appear as follows.
   The screen shows an example for MAINT. OUTPUT.



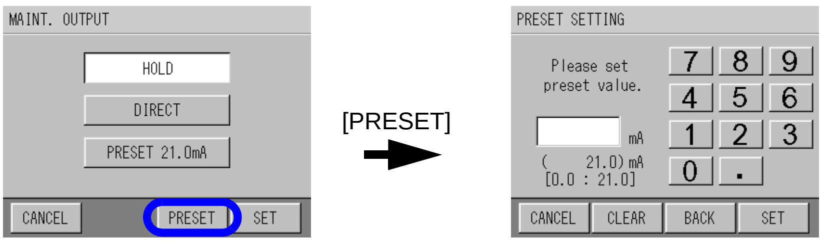
2. **Select the setting item, and press [SET]**

Setting item	Description	Initial setting
HOLD	Outputs the adjacent readings while holding. If there are no adjacent readings when the power is ON, the latest readings before the power OFF are held.	-
DIRECT	Outputs the measured readings directly.	-
PRESET	Outputs the preset values. In this case, the log data is the reading equivalent to the setting output value.	During maintenance and alarm: 21.0 mA During warm-up: 4.0 mA

The screen returns back to the OUTPUT SETTING screen.
Output settings during operation are complete.

★ When setting to PRESET:

1. **Press the [PRESET] button on the lower right of screen.**
   The PRESET screen for each item appears.



2. **Enter the preset value with the numeric keypad, and press [SET],**
   Input range: 0.0 mA to 21.0 mA

Note

When it is preset, the log data also becomes the preset value. The value corresponds to the following calculating formula.
When the analog output level is 4 mA to 20 mA:
Full scale of measurement value × (preset value – 4) / 16
When the analog output level is 0 mA to 16 mA:
Full scale of measurement value × preset value / 16

Tip

If the setting is changed while holding the readings, the setting change is enabled after the hold is released.

5.4.4 Output Condition Setting

[MAINT.] [SETTING] [OUTPUT CONDITION SETTING]

MEAS. screen → MAINTENANCE screen → SETTING 1/2 screen → OUTPUT CONDITION SETTING screen

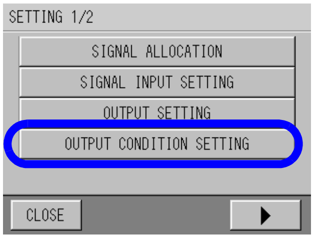
Refer to " 5.4.2 Signal Input Setting " (page 36).

Here enables the output conditions setting (such as the alarm discrimination time and the minus display).

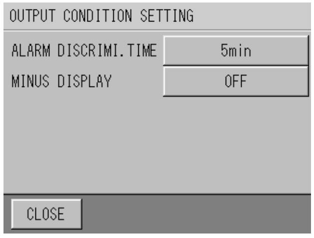
For normal usage, default setting can be used (no additional setting is required).

1. **Press [OUTPUT CONDITION SETTING] on the SETTING 1/2 screen.**

The OUTPUT CONDITION SETTING screen appears.



[OUTPUT CONDITION SETTING] →



2. **Press the button for ALARM DISCRIMI. TIME to change the discrimination time of the alarm output, and enter the value of setting range. Press the button for MINUS DISPLAY to activate/deactivate the minus display.**

Setting item	Description	Initial setting
ALARM DIS-CRIMI. TIME	Sets the alarm delay time to ignore the spike alarm. When the time exceeds the ALARM DISCRIME. TIME, the alarm is triggered and sounds continuously. (Set to "0" minute if you want to occur the alarm immediately.) Setting range: 0 to 99 minutes	5min
MINUS DIS-PLAY	Sets whether to display the measurement value. Setting range: OFF or ON	OFF

The screen returns back to the OUTPUT CONDITION SETTING screen.

The output condition setting is complete.

5.4.5 Time Adjustment

[MAINT.] [SETTING]

MEAS. screen → MAINTENANCE screen → SETTING 1/2 screen

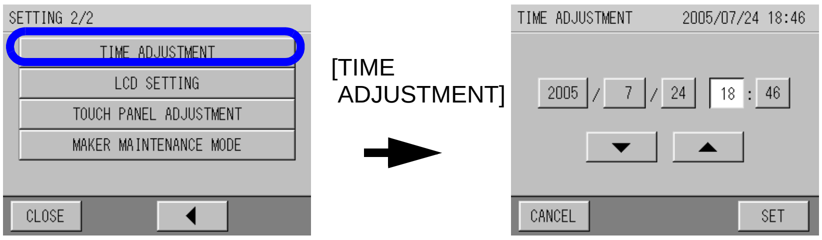
[▶] SETTING 2/2 screen

Refer to " 5.3 Maintenance Screen Display " (page 33).

The clock can be set here.

1. **Press the [TIME ADJUSTMENT] button on the SETTING 2/2 screen.**

   The TIME ADJUSTMENT screen appears.



2. **Select the items from year/month/date/hour/minute to be changed and enter numbers using the [▲] and [▼]buttons.**
   After entering the time, press [SET].

   The screen returns back to the SETTING 2/2 screen.
   Time adjustment is complete.

5.4.6 LCD Setting

[MAINT.] [SETTING]

MEAS. screen → MAINTENANCE screen → SETTING 1/2 screen

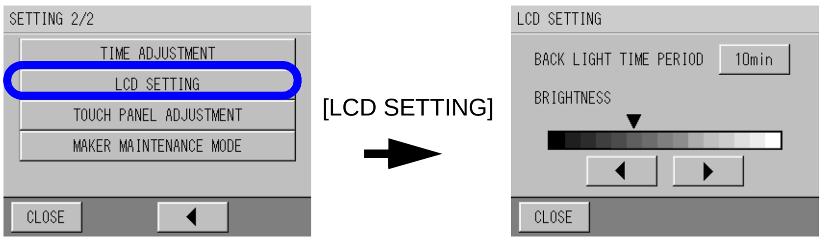
[▶] SETTING 2/2 screen

Refer to " 5.3 Maintenance Screen Display " (page 33).

The condition setting of the LCD can be set here.

1. **Press [LCD SETTING] on the SETTING 2/2 screen.**

   The LCD SETTING screen appears.



The LCD setting contains the following items.

Setting item	Description	Initial setting
BACKLIGHT TIME PERIOD Setting	Sets the auto-off timer of the LCD back-light(after the last touch). Setting range: 10 min, 20 min, 30 min OFF (backlight is always ON)	10 min
BRIGHTNESS Adjustment	Adjusts the brightness of the backlight.	-

2. **Change the condition to be set, and the setting is complete.**

Tip

If no button is pressed for about 30 minutes on displays in the LCD SETTING screen, the display returns back to the previous setting screen.

5.4.7 Touch Panel Adjustment

MEAS. screen → [MAINT.] → MAINTENANCE screen → [SETTING] → SETTING 1/2 screen → [▶] → SETTING 2/2 screen

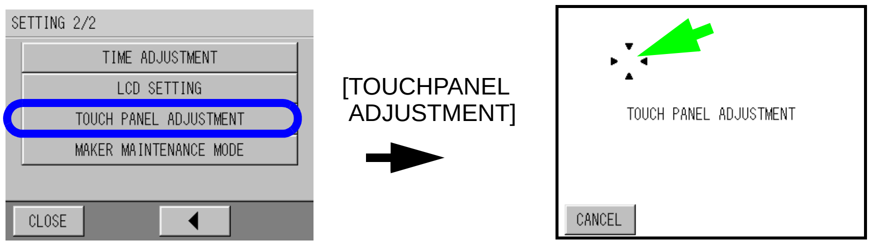
Refer to " 5.3 Maintenance Screen Display " (page 33).

The touch panel can be adjusted here.

It is not necessary to adjust the touch panel for normal usage. When the touch panel functions abnormally, such as when pressing a button activates a different button or when the location is misaligned, adjust the touch panel.

1. **Press [TOUCH PANEL ADJUSTMENT] on the SETTING 2/2 screen.**

The TOUCH PANEL ADJUSTMENT screen appears.



2. **Press the area indicated by the arrow.**

After completing the adjustment, the screen returns back to the SETTING 2/2 screen.

5.4.8 Maker Maintenance Mode

As this mode is for the HORIBA service personnel, it is not necessary to touch this setting.

Note

As the Maker Maintenance Mode is the mode to set the inside adjustment value, do not enter this mode. Doing so may cause a malfunction.

5.5 Maintenance - Measurement Setting

Measurement settings contains the following items.

Setting item	Setting item	Description	Initial setting
UV	MEAS. RANGE	Sets the measurement range. Setting range: (when the unit is "Abs") 0.1000 to 5.0000 (when the unit is "m^{-1}") 10.00 to 500.00	1.000 Abs
	DECIMAL PLACE	Sets the decimal place of the measurement value. Setting range: (when the unit is "Abs") 3 and 4 (when the unit is "m^{-1}") 1 and 2	3
	UNIT	Sets the measurement unit of UV, VIS, and UV-αVIS. Setting range: Abs and m^{-1}	Abs
VIS	MEAS. RANGE	Sets the measurement range. Setting range: (when the unit is "Abs") 0.1000 to 5.0000 (when the unit is "m^{-1}") 10.00 to 500.00	1.000 Abs
	DECIMAL PLACE	Sets the decimal place of the measurement value. Setting range: (when the unit is "Abs") 3 and 4 (when the unit is "m-1") 1 and 2	3
	TSS COR. FAC.	Sets the TSS correction factor. Setting range: 0.000 to 9.999	0.000
COD	MEAS. RANGE	Sets the measurement range. Setting range: 0.100 to 9999.999	1.0 mg/L
	DECIMAL PLACE	Sets the decimal place of the measurement value. Setting range: 0 to 3	1
	y = a + bx	Sets the COD conversion factor a and b by calculating from UV-αVIS. Setting range: a –9999.999 to 9999.999 b 0.001 to 9999.999 Up to five pairs of the COD conversion factors "a" and "b" are available.	a = 0.0 b = 1.000
	MAX ALARM VALUE	Sets the upper limit of COD alarm value. Setting range: –9999.999 to 9999.999	1.0
	MIN ALARM VALUE	Sets the lower limit of COD alarm value. Setting range: –9999.999 to 9999.999	0.0
TSS	MEAS. RANGE	Sets the measurement range. Setting range: –9999.999 to 9999.999	1.0 mg/L
	DECIMAL PLACE	Sets the decimal place of the measurement value. Setting range: –9999.999 to 9999.999	1
	y = a + bx	Sets the TSS Meas. correction factor "a" and "b". Correction is possible when the readings differ from the actual value due to the various causes even after calibration. Setting range: a –9999.999 to 9999.999 b 0.001 to 9999.999	a = 0.0 b = 1.000
	MAX ALARM VALUE	Sets the upper limit of TSS alarm value. Setting range: –9999.999 to 9999.999	1
	MIN ALARM VALUE	Sets the lower limit of TSS alarm value. Setting range: –9999.999 to 9999.999	0

● Setting Screen Display for each Measuring Component

[MAINT.] [MEAS. SETTING]

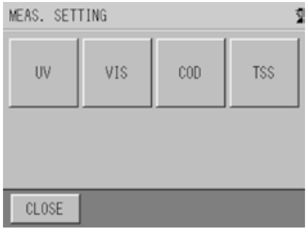
MEAS. screen → MAINTENANCE screen → MEAS. SETTING screen

Refer to " 5.3 Maintenance Screen Display " (page 33).

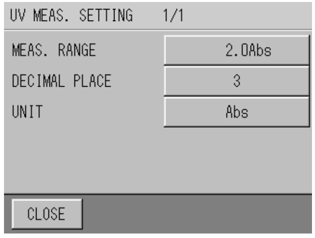
1. **Press a measuring component button on the MEAS. SETTING screen. (Press [UV] or [COD] for the following example.)**

The setting screen for each measuring component appears.

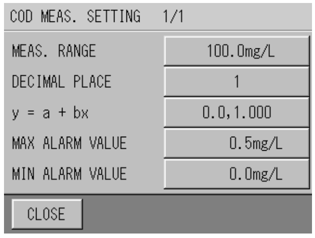
The screens show examples for UV and COD.



[UV]



[COD]



5.5.1 Measuring Range Setting for each Component

[MAINT.] [MEAS.SETTING] [UV][VIS][COD][TSS]

MEAS. screen → MAINTENANCE screen → MEAS. SETTING screen → MEAS. SETTING for each measuring component screen

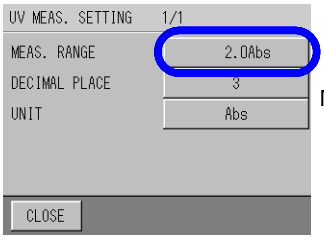
Refer to “ n Setting Screen Display for each Measuring Component ” (page 43).

The measuring range for each measuring component is set here.

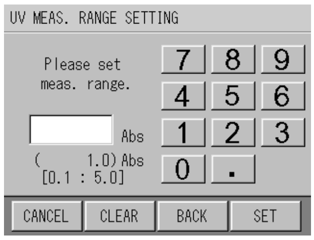
1. **Press the button for MEAS. RANGE on the MEAS. SETTING for each measuring component screen.**

   The MEAS. RANGE SETTING for each measuring component screen appears.

   The screen shows an example for UV.



MEAS. RANGE



2. **Enter the numeric setting range and press [SET].**

   The screen returns back to the MEAS. SETTING for each measuring component screen.

   The measuring range setting for each measuring component is complete.

   Perform the same procedure to set the measuring range for the other components.

Note

- **Set the measuring range so as not to exceed the full scale of the measurement value (level of approximately 80%).**
- **If the measuring range is changed while measuring, the measurement value may go wrong. In this case, perform the calibration using a matching calibration solution.**
- **The measuring range for UV-αVIS is the same as that for UV.**

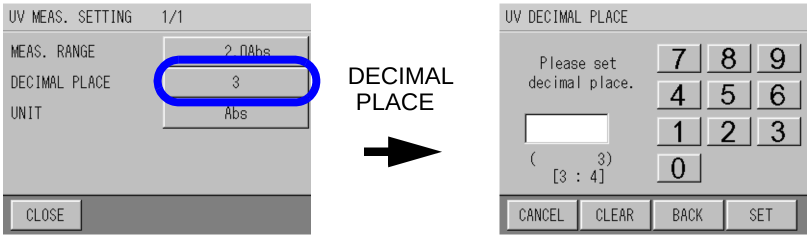
5.5.2 Setting of Decimal Place for each Component

[MAINT.]		[MEAS.SETTING]		[UV][VIS][COD][TSS]		
MEAS. screen	→	MAINTENANCE screen	→	MEAS. SETTING screen	→	MEAS. SETTING for each measuring component screen

Refer to " n Setting Screen Display for each Measuring Component " (page 43).

The decimal place for each measuring component is set here.

1. **Press the button for DECIMAL PLACE on the MEAS. SETTING for each measuring component screen.**
   The DECIMAL PLACE screen appears.
   The screen shows an example for UV.



2. **Enter a numeric value for the decimal place and press [SET].**
   The screen returns back to the MEAS. SETTING for each measuring component screen.
   The decimal place setting for each measuring component is complete.
   Perform the same procedure to set the measuring range for the other components.

Note

The decimal place for UV-αVIS is the same as that for UV.

5.5.3 Unit Setting for UV, VIS, and UV-αVIS

MEAS. screen → [MAINT.] → MAINTENANCE screen → [MEAS.SETTING] → MEAS. SETTING screen → [UV] → UV MEAS. SETTING screen

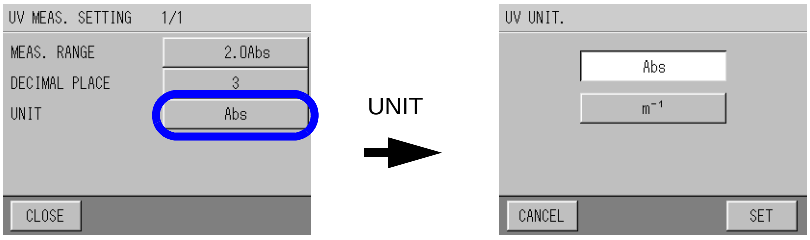
Refer to " n Setting Screen Display for each Measuring Component " (page 43).

The units for UV, VIS, and UV-αVIS are set here. Select "Abs" when indicating UV, VIS, and UV-αVIS in a cell length conversion of 10 mm, and select "m^{-1}" when indicating UV, VIS, and UV-αVIS in a cell length conversion of 1 m (the indication of "m^{-1}" is the hundredfold of the indication of "Abs") .

When the unit is changed, the range that can be measured simultaneously is also changed.

1. **Press the button for UNIT on the UV MEAS. SETTING screen.**

The UV UNIT screen appears.



2. **Select the unit and press [SET].**

The screen returns back to the UV MEAS. SETTING screen.

Unit setting for UV, VIS, and UV-αVIS is complete.

Note

- **In accordance with the change of unit settings for UV, unit setting for VIS and UV-αVIS are also changed.**
- **In accordance with the unit change, the decimal place and measuring range indication are also changed. However, the value b of "y = a + bx" for COD and TSS and span calibration value for TSS are not changed. Perform settings and calibration again.**
- **If the unit is changed while measuring, digits of saved data may be lost. Make sure to change the unit at the beginning of operation.**

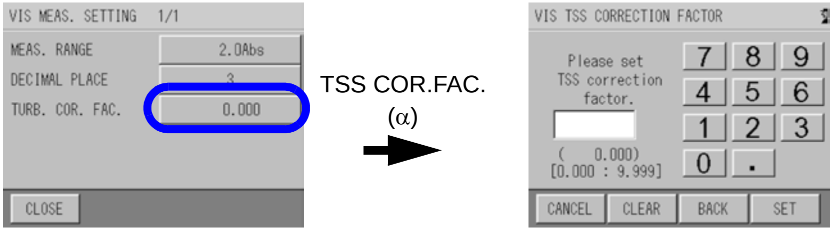
5.5.4 Setting of TSS Correction Factor

MEAS. screen →[MAINT.]→ MAINTENANCE screen →[MEAS.SETTING]→ MEAS. SETTING screen →[VIS]→ VIS MEAS. SETTING screen

Refer to " n Setting Screen Display for each Measuring Component " (page 43).

The TSS correction factor "α" used for the calculation of UV-αVIS is set here.

1. **Press the button for TSS COR. FAC. on the VIS MEAS. SETTING screen.**
   The VIS TSS CORRECTION FACTOR screen appears.



2. **Enter the numeric TSS correction factor "α" and press [SET].**
   The screen returns back to the VIS MEAS. SETTING screen.
   Unit setting for TSS correction factor is complete.

Note

Enter the TSS correction factor "α" when the sample water contains much TSS to cause a poor correlation between the UV value and COD value by manual analysis.

5.5.5 Setting of COD Conversion Factor and TSS Meas. Correction Factor

[MAINT.] [MEAS. SETTING] [COD] [TSS]

MEAS. screen → MAINTENANCE screen → MEAS. SETTING screen → COD MEAS. SETTING screen / TSS MEAS. SETTING screen

Refer to " ● Setting Screen Display for each Measuring Component" (page 43)

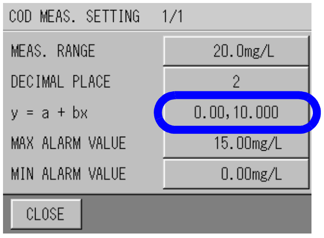
Here describes how to set the COD conversion factor from UV-αVIS and TSS Meas. correction factor.

Reference

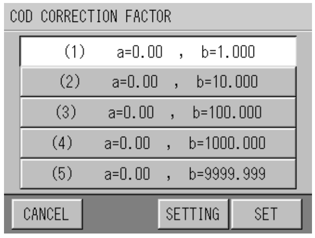
" 9.5 How to Calculate the COD Conversion Factor " (page 101)

1. **Press the button for y = a + bx on the COD or TSS MEAS. SETTING screen.**
   - In case of COD, up to five pairs of the COD conversion factors "a" and "b" are availabl. The screen for selecting the factor appears.
   - In case of TSS, the screen for selecting the factor "a" or "b" appears.

   The screen shows an example for COD.

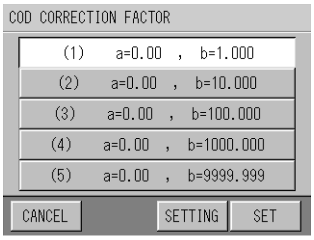


[a=x.x b=x.xxx]

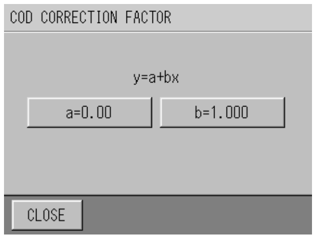


2. **Select the pair to set the factor, and press [SETTING].**

   The screen for selecting the factor "a" or "b" appears.

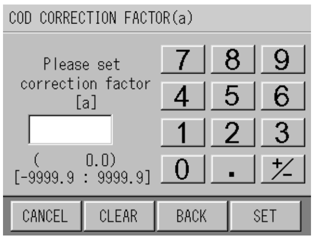


【SETTING】



3. **Press [a=x.x].**

   The screen for setting the factor "a" appears.



4. Enter the value of "a" with the numeric keypad, and press [SET].

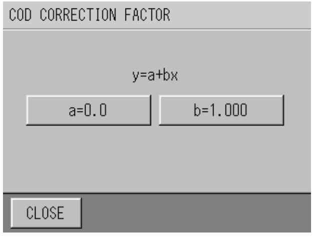
Input range: –9999.9 to 9999.9

The described above are provided the sample when the decimal place is set to 1 digit.

Tip

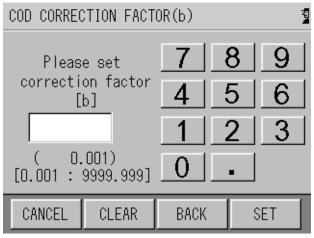
[CLEAR]: Clears all input numeric values.
[BACK]: Clears the rightmost number.

The screen returns back to the screen for selecting the factor “a” or “b”.



5. Press[b=x.xxx].

The screen for setting the factor “b” appears.



6. Enter the value of "b" with the numeric keypad, and press[SET].

Input range: 0.1 to 9999.9

The described above are provided the sample when the decimal place is set to 1 digit.

Tip

[CLEAR]：Clears all input numeric values.
[BACK]：Clears the rightmost number.

The screen returns back to the screen for selecting the factor “a” or “b”.

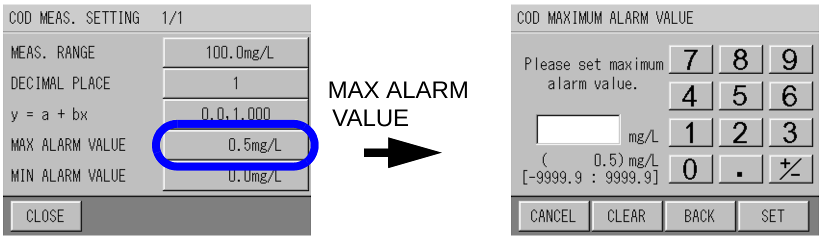
The setting of the COD conversion factor and TSS Meas. correction factor are complete.

5.5.6 Setting of MAX Alarm Value for COD and TSS

MEAS. screen → [MAINT.] → MAINTENANCE screen → [MEAS. SETTING] → MEAS. SETTING screen → [COD] [TSS] → COD MEAS. SETTING screen / TSS MEAS. SETTING screen

Refer to “ n Setting Screen Display for each Measuring Component ” (page 43)

1. **Here describes how to set the upper limits of the COD and TSS alarm values.**
   Press the button of the measuring component on the COD or TSS MEAS. SETTING screen.
   The MEAS. SETTING screen for each measuring component appears.
2. **Press the button for MAX ALARM VALUE.**
   The MAXIMUM ALARM VALUE screen for each measuring component appears.
   The screen shows an example of MAX ALARM VALUE for COD.



3. **Enter the upper limit value with the numeric keypad, and press [SET].**
   Input range: –9999.9 to 9999.9
   The described above are provided the sample when the decimal place is set to 1 digit.

Tip

[CLEAR] : **Clears all input numeric values.**
[BACK] : **Clears the rightmost number.**

The screen returns back to the MEAS. SETTING screen for each measuring component.

5.5.7 Setting of MIN Alarm Value for COD and TSS

[MAINT.] [MEAS. SETTING] [COD] [TSS]

MEAS. screen → MAINTENANCE screen → MEAS. SETTING screen → COD MEAS. SETTING screen / TSS MEAS. SETTING screen

Refer to " n Setting Screen Display for each Measuring Component " (page 43)

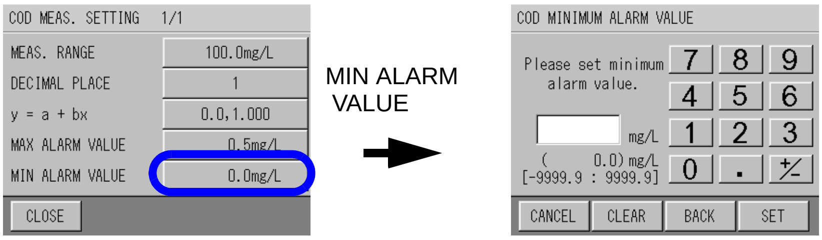
Here describes how to set the lower limits of the COD and TSS alarm values.

1. **Press the button of the measuring component on the COD or TSS MEAS. SETTING screen.**

   The MEAS. SETTING screen for each measuring component appears.

2. **Press the button for MIN ALARM VALUE.**

   The MINIMUM ALARM VALUE screen for each measuring component appears.
   The screen shows an example of MINIMUM ALARM VALUE for COD.



3. **Enter the lower limit value with the numeric keypad, and press [SET],**

   Input range: –9999.9 to 9999.9
   he described above are provided the sample when the decimal place is set to 1 digit.

Tip

[CLEAR]：**Clears all input numeric values.**
[BACK]：**Clears the rightmost number.**

The screen returns back to the MEAS. SETTING screen for each measuring component.

5.6 Maintenance - Action

This function performs operations using the operating buttons.

Action	Description
WIPER MOTOR	Checks the ON/OFF operation of the rotary cell wiper motor.
UV LAMP	Checks the ON/OFF operation of the UV lamp for light.

5.6.1 Action

[MAINT.] [ACTION]

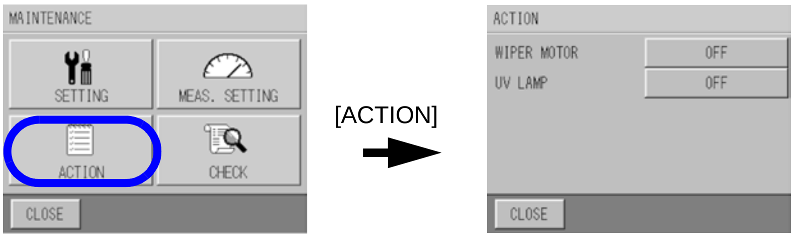
MEAS. screen → MAINTENANCE screen → ACTION screen

Refer to “5.3 Maintenance Screen Display” (page 33).

Here describes the execution methods for wiper motor and UV lamp.

1. **Press the [ACTION] button on the MAINTENANCE screen.**

The ACTION screen appears.



2. **Every time the item to be activated is pressed, the value switches between ON and OFF.**

The action is activated. Check whether the value has been activated normally.

3. **Press [CLOSE] to return back to the MAINTENANCE screen.**

Press [CLOSE] to return back to the ON condition automatically.

Note

- **When checking the action, do not touch the cell part, and do not directly look at the UV lamp; it is dangerous.**
- **When performing the action, the motor and lamp are stopped and abnormal measurement values may be recorded.**

5.7 Maintenance - Check

Items to be checked are as follows.

Unit information and ID setting check	Checks the unit model and ID etc, and sets the ID. The ID number is used as the folder name of the data during serial communication and transfer to the CF card.
External input and output check	Checks the operation of the contact input and output.
Analog input value check	Checks the analog signal value for measurement transferred from the analyzer unit. Can be used to simply check whether it is activated or not (for service personnel).
Analog output transmission check	Checks the analog output value.

5.7.1 Unit Information

[MAINT.] [CHECK]

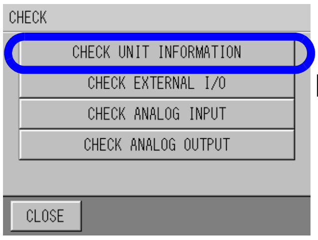
MEAS. screen → MAINTENANCE screen → CHECK screen

Refer to “ 5.3 Maintenance Screen Display ” (page 33).

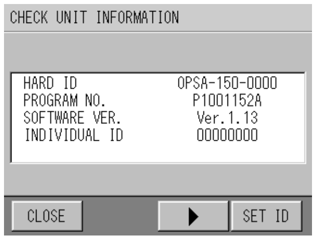
1. **Press the [CHECK UNIT INFORMATION] on the CHECK screen.**

The CHECK UNIT INFORMATION screen appears.

Information about the model number and individual ID are displayed.



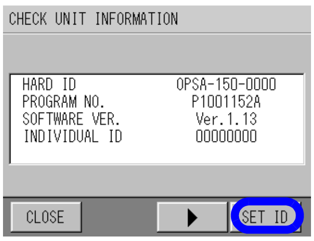
[CHECK UNIT INFORMATION]



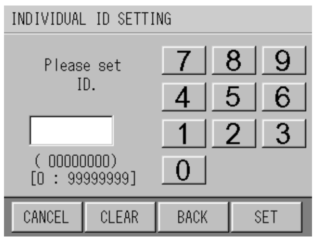
5.7.2 Individual ID Setting

1. **Press the [SET ID]on the CHECK UNIT INFORMATION screen.**

The INDIVIDUAL ID SETTING screen appears.



[SET ID]



2. **Enter the ID number with the numeric keypad, and press [SET].**

Input range: 00000000 to 99999999

Tip

[CLEAR] : **Clears all input numeric values.**

[BACK] : **Clears the rightmost number.**

The screen returns back to CHECK UNIT INFORMATION screen.

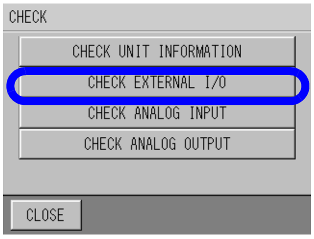
5.7.3 External Input/Output Check

[MAINT.] [CHECK]
MEAS. screen → MAINTENANCE screen → CHECK screen
Refer to " 5.3 Maintenance Screen Display " (page 33).

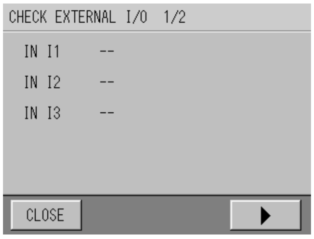
Check the input and output of the external unit.

1. **Press the [CHECK EXTERNAL I/O] on the CHECK screen.**

The CHECK EXTERNAL I/O 1/2 screen appears.
The following can be checked.
IN I1: Float switch input from sample
IN I2: External input check of clock set
IN I3: Switch input check during maintenance



[CHECK EXTERNAL I/O]



2. **Contact output can be checked on the CHECK EXTERNAL I/O 2/2 screen**

O1 to O4: Allocated contact
O5: Power OFF (only O5 is for action, contact is OFF when power is ON, and contact is ON when power is OFF)
O6: During maintenance (check when the switch is OFF during maintenance)

Note

Remove the wirings of the unit before checking the contact action (as it may affect the external measuring instruments).

5.7.4 Check Analog Input

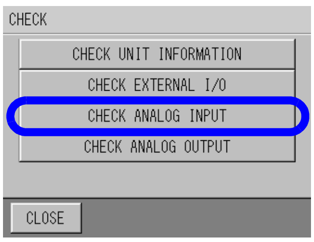
This screen is the check screen for the service personnel.

[MAINT.] [CHECK]
MEAS. screen → MAINTENANCE screen → CHECK screen
Refer to " 5.3 Maintenance Screen Display " (page 33).

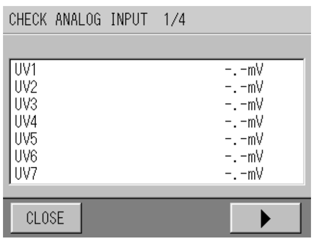
The sensor analog input transmitted from the analyzer unit is displayed.

1. **Press the [CHECK ANALOG INPUT] button on the CHECK screen.**

The CHECK ANALOG INPUT screen appears.
The analog input from each sensor is displayed.



[CHECK ANALOG INPUT]



5.7.5 Check Analog Output

[MAINT.] [CHECK]

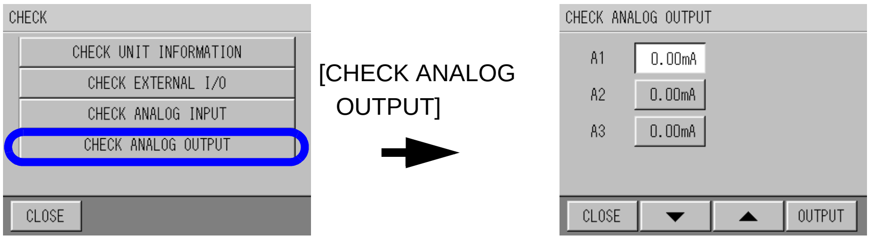
MEAS. screen → MAINTENANCE screen → CHECK screen

Refer to “ 5.3 Maintenance Screen Display ” (page 33).

Tip

If no button is pressed for about 30 minutes on the CHECK ANALOG OUTPUT screen, the display returns back to the previous setting screen.

1. **Press the [CHECK ANALOG OUTPUT] on the CHECK screen.**
   The CHECK ANALOG OUTPUT screen appears.



2. **Select the items to output by pressing the button ([x.xxmA]), change the number by using the [▼] and [▲] buttons, and press [OUTPUT].**
   Output value: 0.00/4.00/8.00/12.00/16.00/20.00 mA
3. **Check the output with an ammeter.**

Reference

“ 6.2 Analog Output ” (page 69)

5.8 Data Check

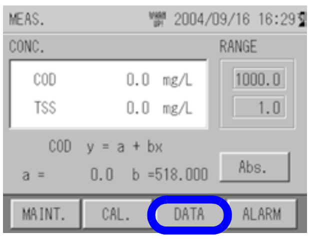
Items for the data check are as follows.

Items for data check	Description
LOG DATA check	Checks the previous data readings numerically. Approximately11 days can be saved for the value of 1 minute and approximately 13 months can be saved for the value of 1 hour.
LOG DATA deletion	Deletes all of the log data.
GRAPH display	Checks the previous readings of data using a graph.
CAL. REPORT check	Checks up to 20 history records by measurement component and zero/ span.
CAL. REPORT deletion	Deletes all of the calibration report.
ALARM REPORT check	Checks up to 511 alarm history records.
ALARM REPORT deletion	Deletes all of the alarm history.
CF CARD transfer	Transfers the data to the CF card.
CF CARD initialization	Initializes the CF card.

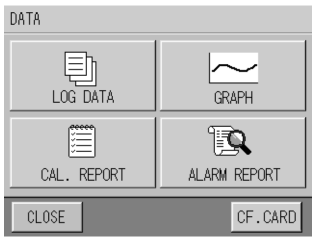
● Data Screen Display

1. **Press the [DATA] on the MEAS. screen.**

The DATA screen appears.



[DATA]



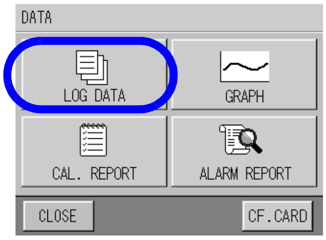
MEAS. screen

5.8.1 Log Data Check

[DATA]
MEAS. screen → DATA screen
Refer to “ n Data Screen Display ” (page 56)

Displays the measurement value of 1 minute and 1 hour memorized by the measurement.

1. **Press the [LOG DATA] on the DATA screen.**
The LOG DATA screen appears.
The time of log data and readings of each measurement component are displayed.



[LOG DATA]

(MIN): data when the number of seconds is "00". Maximum data: approximately 11 days
(HOUR): data when the number of minutes is "00". Maximum data: approximately 13 months

The following marks are displayed for the data during the action.
* If multiple actions are overlapping, they are displayed in numerical order.

① MAINT During maintenance
② WARM During warm-up
③ ALARM During alarm

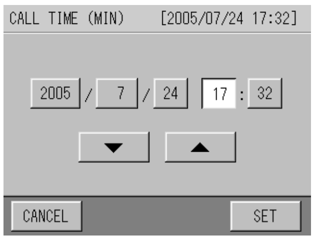
Tip

[DELETE] : **Deletes the all of the log data.**
[◀][▶] : **Switches to the previous and next data.**
[CHANGE] : **Switches between (MIN) and (HOUR) LOG DATA.**
[CALL] : **The data at a specified time can be viewed.**

View the data at a specified time

1. **Press [CALL].**
The CALL TIME (MIN/HOUR) screen appears.

[CALL]

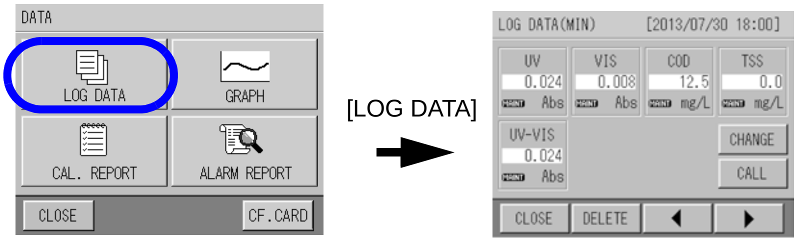


2. **Select the hour to be changed and set the number by using [▲] and [▼]. After entering the number, press [SET].**
The LOG DATA screen with the specified date and time appears.

5.8.2 Log Data Deletion

[DATA]
MEAS. screen → DATA screen
Refer to “ n Data Screen Display ” (page 56)

1. **Press the [[LOG DATA] on the DATA screen.**
The LOG DATA screen appears.



2. **Press the [DELETE] on the LOG DATA screen.**
The "Delete all data?" screen appears.

3. **Press [YES].**
The screen returns back to the LOG DATA screen.
Log data deletion is complete.

Tip

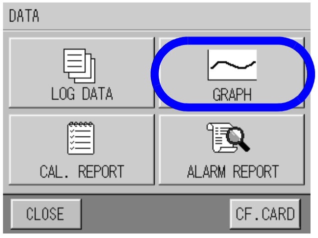
When executing a deletion on the LOG DATA (MIN) screen: all (MIN) log data is deleted.
When executing a deletion on the LOG DATA (HOUR) screen: all (HOUR) log data is deleted.

5.8.3 Graph Display

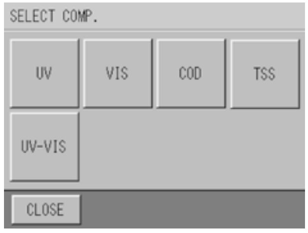
[DATA]
MEAS. screen → DATA screen
Refer to “ n Data Screen Display ” (page 56)

1. **Press the [GRAPH] on the DATA screen.**

The SELECT COMP. screen appears.
The screens show examples for UV.

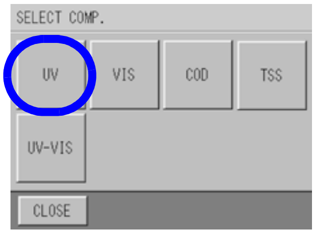


[GRAPH]

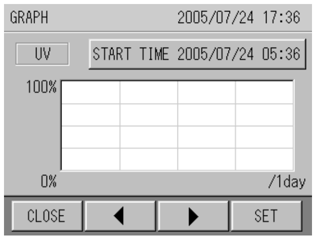


2. **Press the measurement component to be checked.**

The GRAPH screen appears.



[UV]



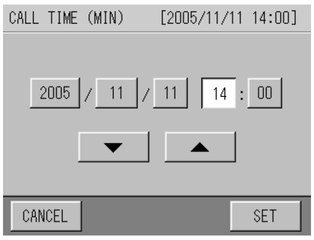
Tip

[START TIME (DATE)]: Changes the start time of the graph.
[◀][▶]: Switches to the previous and next data.
[SET]: Changes the range of vertical and horizontal axes

● Changing the start time of the graph

1. **Press the [START TIME (DATE)].**

The CALL TIME (MIN/DATE) screen appears.



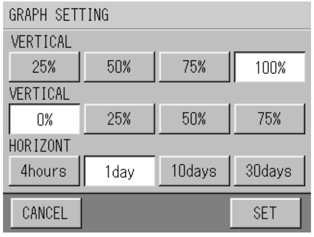
2. **Select the hour to be changed and set the number by using [▲] and [[▼]. After entering the number, press [SET].**

The screen returns back to the GRAPH screen.

● Changing the range of vertical and horizontal axes

1. **Press the [SET] button.**

The [GRAPH SETTING] screen appears.



2. **Select the range from VERTICAL/VERTICAL/HORIZONE, and press [SET].**

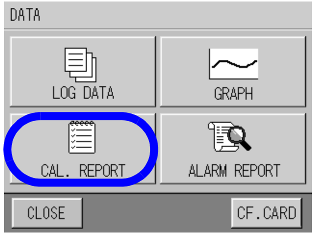
VERTICAL (UPPER)/ VERTICAL (LOWER)	% indicates the proportion of the graph range in relation to the measurement range.
HORIZONE	"4 hours/1 day" indicate the (MIN) data. "10 days/30 days" indicate the (HOUR) data.

5.8.4 Calibration Report Check

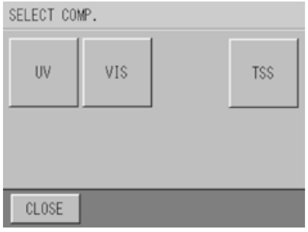
[DATA]
MEAS. screen → DATA screen
Refer to " n Data Screen Display " (page 56)

The calibration report can be checked here.

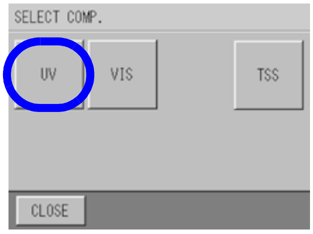
1. **Press the CAL. REPORT] on the DATA screen.**
   The SELECT COMP. screen appears.



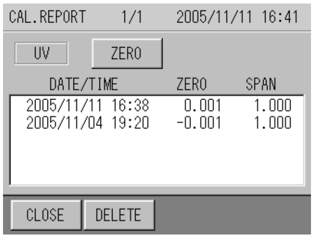
[CAL. REPORT] →



2. **Select the measurement component to be checked.**



[UV] →



The CAL. REPORT screen appears.
Up to 20 calibration dates and zero/span factors are displayed by measurement component and by zero/span in chronological order from the latest date.
The screens show examples for UV.

Reference

" ● About the Calibration coefficient" (page 14)

Tip

[ZERO] /[SPAN]: Switches between ZERO CAL. REPORT and SPAN CAL. REPORT.
[◀] , [▶]: Switches to the previous and next data.
[DELETE]: Deletes all of the calibration report.
When changing the measurement component to display, press [◀] to go back to the SELECT COMP. screen.

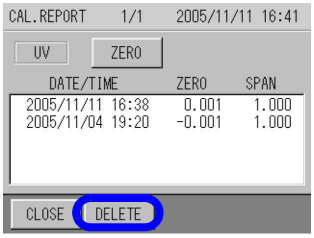
5.8.5 Calibration Report Deletion

[DATA]
MEAS. screen → DATA screen → CAL. REPORT screen

Refer to “ 5.8.4 Calibration Report Check ” (page 61)

1. **Press the[DELETE] on the CAL. REPORT screen.**

The CAL. REPORT screen appears.



[DELETE] →

2. **Press [YES].**

The screen returns back to the CAL. REPORT screen. Calibration report deletion is complete.

Tip

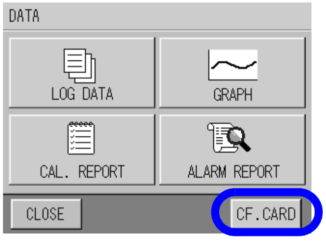
Though all of the calibration report is deleted, the latest calibration value is memorized.

5.8.6 CF Card Transfer

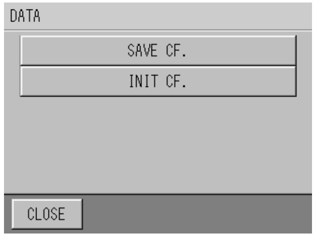
[DATA]
MEAS. screen → DATA screen
Refer to " n Data Screen Display " (page 56)

Data can be transferred to a CF card.

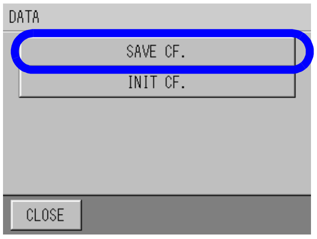
1. **Press the [CF.CARD] on the DATA screen.**
The CF CARD menu screen appears.



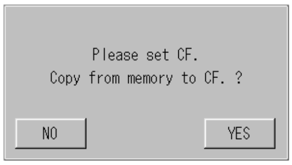
[CF.CARD]



2. **Press [SAVE CF.].**
The CF card transfer screen appears.



[SAVE CF.]



3. **Confirm that the CF card is inserted, and press [YES].**
The transferring screen appears.
When the transfer is complete, the screen returns back to the CF CARD menu screen.

Note

Do not remove a CF card from the card slot during data transfer.
Otherwise the data of CF card may be unreadable because it is destroyed.

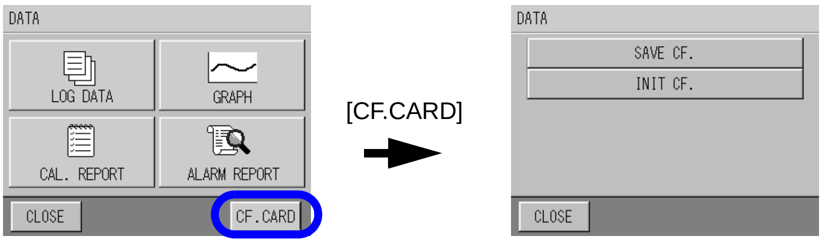
5.8.7 CF Card Initialization

[DATA]
MEAS. screen → DATA screen
Refer to " n Data Screen Display " (page 56)

The CF card can be initialized here.

1. **Press the [CF.CARD] on the DATA screen.**
The CF CARD menu screen appears.

2. **Press [INIT CF.].**
The screen to initialize the CF card appears.



3. **Confirm that the CF card is inserted, and press [YES].**
When the initialization is complete, the screen returns to the CF CARD menu screen.

Note

- **When the CF card is initialized, all the data of CF card is deleted. Make sure to save the contents before executing initialization.**
- **Execute initialization of the CF card using this unit.**
- **The file system of this unit is FAT. FAT32 and NTFS are not accepted.**
- **When the initialization is failed, once remove the CF card from the card slot. Insert the CF card properly, then execute initialization again.**

5.9 Alarm

An alarm is triggered when a malfunction occurs in the unit or the measurement value exceeds the upper or lower limit.

Reference

“ 8.1 Table of Status and Operation ” (page 91)

Items regarding alarm are as follows.

Items of alarm	Description
Alarm Check (5.9.1　page 65)	Performs a check during the alarm.
Alarm Stop (5.9.2　page 66)	Stops all alarms during the alarm.
Alarm History Check (5.9.3　page 67)	Checks the alarm history of up to 511 records.
Alarm History Deletion (5.9.4　page 65)	Deletes all of the alarm history.

Note

As this unit calculates values using one digit under the least significant digit indicated as readings or setting values, an alarm may not be triggered even if the displayed value is the same value set for alarm.

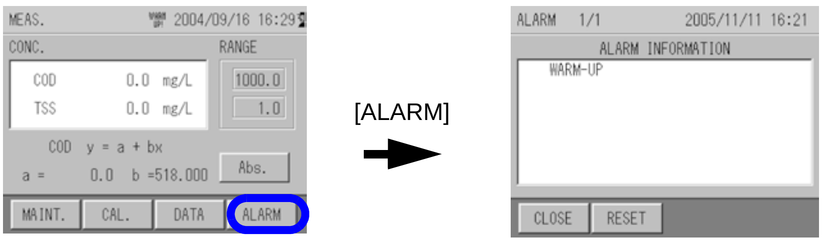
5.9.1 Alarm Check

The instant alarm can be checked here.

1. **Press the [ALARM] on the MEAS. screen (main screen).**

The ALARM screen appears.

The alarm information is displayed.



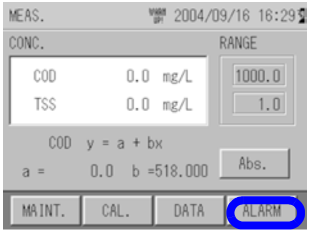
Tip

[RESET]: Stops the alarm.
If an instant alarm is not triggered, the ALARM button is not displayed.

5.9.2 Alarm Stop

1. **Press the [ALARM] on the MEAS. screen (main screen).**

The ALARM screen appears, and the alarm information is displayed.



[ALARM]

2. **Press [RESET].**

The alarm stop screen appears.

3. **Press [YES].**

The screen returns back to the ALARM screen.

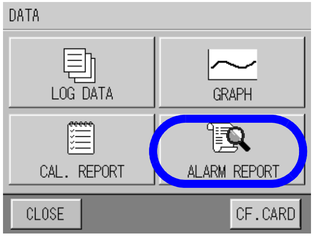
Alarm stop is complete.

5.9.3 Alarm History Check

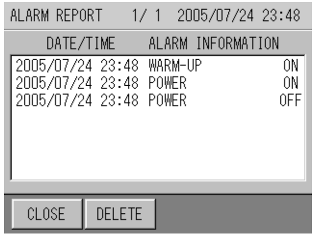
[DATA]
MEAS. screen → DATA screen
Refer to “ n Data Screen Display ” (page 56)

The history of past alarms can be checked here.

1. **Press the [ALARM REPORT] on the DATA screen.**
   The ALARM REPORT screen appears.
   Alarm information, date/time, and ON/OFF are displayed.
   If "ON" is displayed, the information when the alarm was triggered is shown, and if "OFF" is displayed, the information when the alarm was released is shown.



[ALARM REPORT]



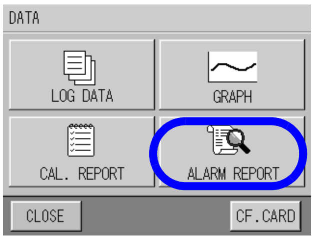
Tip

[DELETE]: Deletes the alarm history.

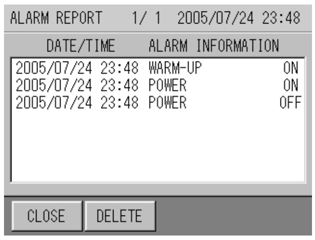
5.9.4 Alarm History Deletion

[DATA]
MEAS. screen → DATA screen
Refer to “ n Data Screen Display ” (page 56)

1. **Press the [ALARM REPORT] on the DATA screen.**
   The ALARM REPORT screen appears.



[ALARM REPORT]



2. **Press [DELETE].**
   The alarm history deletion screen appears.
3. **Press [YES].**
   The screen returns back to the ALARM REPORT screen.
   The alarm history deletion is complete.

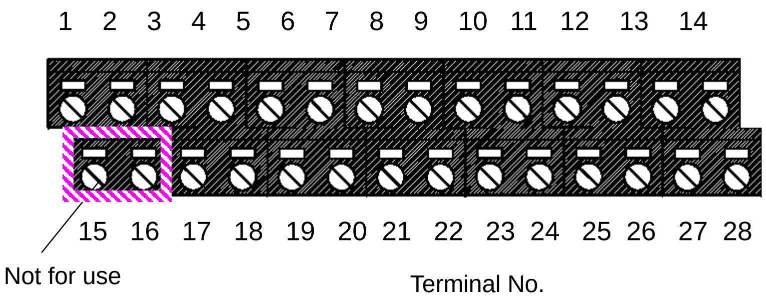
6 External Input and Output

This unit contains the terminals for contact input/output, analog output, and RS-232C input/output. These terminals should be used when the status input, alarm signal output, measurement value signal output, and remote signal input are required.

6.1 Terminal Diagram of Input and Output

The input/output terminal diagram is shown below.

Input/output terminals are the green terminals under the control unit.



Terminal No.

1	2	3	4	5	6	7	8	9	10	11	12	13	14
TXD	RXD	SG	NC	A-2(+)	A-2(−)	I-1(+)	I-1(−)	O-1(+)	O-1(−)	O-3(+)	O-3(−)	O-5(+)	O-5(−)
RS-232C													

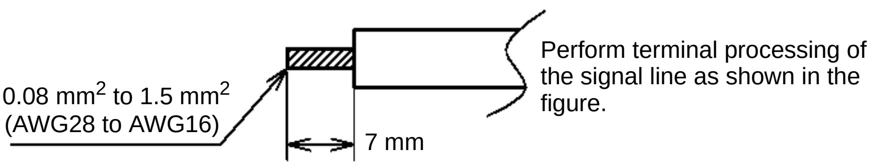
15	16	17	18	19	20	21	22	23	24	25	26	27	28
Not for use		A-1(+)	A-1(−)	A-3(+)	A-3(−)	I-2(+)	I-2(−)	O-2(+)	O-2(−)	O-4(+)	O-4(−)	O-6(+)	O-6(−)
		ANALOG OUT.				CONTACT IN		CONTACT OUT.					

Not for use

Terminal table

Note

Do not connect the signal line to terminals No.15 and 16 as they are connected internally.



Terminal processing of signal line

6.2 Analog Output

This unit outputs the measurement value by analog current output when outputting to the outside.

The analog output contains three outputs, and outputs by selecting three components from the following five components.

The output value is output in conjunction with the full scale of each measurement value.

● Measurement item and analog output range: example of 4 to 20 mA output

Measurement item	Range (example)	4 mA output	20 mA output
UV	1 Abs	0 Abs	1 Abs
VIS	1 Abs	0 Abs	1 Abs
UV-VIS	1 Abs	0 Abs	1 Abs
COD	100 mg/L	0 mg/L	100 mg/L
TSS	100 mg/L	0 mg/L	100 mg/L

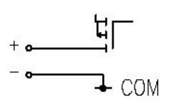
● Measurement item and analog output range: example of 0 to 16 mA output

Measurement item	Range (example)	0 mA output	16 mA output
UV	1 Abs	0 Abs	1 Abs
VIS	1 Abs	0 Abs	1 Abs
UV-VIS	1 Abs	0 Abs	1 Abs
COD	100 mg/L	0 mg/L	100 mg/L
TSS	100 mg/L	0 mg/L	100 mg/L

Allocations for measurement analog output terminals are as follows.

Specifications for analog current output are as follows.

Terminal No.	Output No.	Allocation
17 18	A-1(+) A-1(–)	(Arbitrary setting)
5 6	A-2(+) A-2(–)	(Arbitrary setting)
19 20	A-3(+) A-3(–)	(Arbitrary setting)

Signal type	Input and output circuit	Specifications
Analog signal output		● DC 0 mA to 16 mA/4 mA to 20 mA Current signal output ● Insulated output (common to COM) ● Load resistance: Max. 600 Ω

Reference

" 5.4.1 Signal Allocation " (page 34)

6.3 Contact Input and Output

6.3.1 Contact Output

This unit incorporates the contact output as shown below.

The contact point output contains 6 outputs, and 4 outputs can be set arbitrarily apart from POWER OFF and MAINTENANCE.

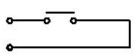
Item	Output No.	Description	Output timing
TOTAL ALARM	Arbitrary setting	Indicates various alarms except POWER OFF.	ON when 1 or more alarms except POWER OFF, MAINTENANCE, and NO SAMPLE have been triggered. (short circuit) OFF when all alarms are stopped except POWER OFF, MAINTENANCE, and NO SAMPLE. (open)
ANALYZE ERR.	Arbitrary setting	Indicates an analyzer malfunction.	ON when an analyzer has malfunctioned. (short circuit) OFF when all the analyzer malfunctions are canceled. (open)
LAMP ERR	Arbitrary setting	Indicates a UV lamp malfunction.	ON when the light intensity of the UV lamp is less than the specified value. (short circuit) OFF when the light intensity of UV lamp is over the specified value. (open)
MOTOR ERR.	Arbitrary setting	Indicates a wiper motor malfunction.	ON when motor rotation is abnormal. (short circuit) OFF when motor rotation is normal. (open)
NO SAMPLE	Arbitrary setting	Indicates there is no sample.	ON when the water level of the overflow tank has decreased and float switch is activated for 5 seconds continuously. (short circuit) OFF when the water level of overflow tank has increased and float switch is canceled for 5 seconds continuously. (open)
MAX COD	Arbitrary setting	Indicates the upper alarm limit for COD.	ON when the status “COD measurement value > COD MAX ALARM VALUE” continues longer than the specified alarm time. (short circuit) OFF when the status “COD measurement value ≤ COD MAX ALARM VALUE” continues longer than the specified alarm time. (open)
MIN COD	Arbitrary setting	Indicates the lower alarm limit for COD.	ON when the status “COD measurement value < COD MIN ALARM VALUE” continues longer than the specified alarm time. (short circuit) OFF when the status “COD measurement value ≥ COD MIN ALARM VALUE” continues longer than the specified alarm time. (open)
MAX TSS	Arbitrary setting	Indicates the upper alarm limit for TSS.	ON when the status “TSS measurement value > TSS MAX ALARM VALUE” continues longer than the specified alarm time. (short circuit) OFF when the status “TSS measurement value ≤ TSS MAX ALARM VALUE” continues longer than the specified alarm time. (open)
MIN TSS	Arbitrary setting	Indicates the lower alarm limit for TSS.	ON when the status “TSS measurement value < TSS MIN ALARM VALUE” continues longer than the specified alarm time. (short circuit) OFF when the status “TSS measurement value ≥ TSS MIN ALARM VALUE” continues longer than the specified alarm time. (open)

Item	Output No.	Description	Output timing
WARM-UP	Arbitrary setting	Indicates the unit is warming up.	ON for 1 hour after turning the power ON. (short circuit) OFF after 1 hour has passed since turning the power ON. (open)
POWER OFF	O5(fixed)	Indicates power OFF.	ON when the power is turned OFF. (short circuit) OFF when power OFF is canceled. (open)
MAINTENANCE	O6(fixed)	Indicates the unit is in maintenance, or calibration mode.	ON during maintenance mode or calibration mode, or when the maintenance switch is ON. (short circuit) OFF when not in-maintenance mode or calibration mode, or when the maintenance switch is OFF. (open)

Allocations for contact output terminals are as follows.

Terminal No.	Output No.	Allocation.
9 10	O-1(+) O-11(–)	(Arbitrary setting)
23 24	O-2(+) O-2(–)	(Arbitrary setting)
11 12	O-3(+) O-3(–)	(Arbitrary setting)
25 26	O-4(+) O-4(–)	(Arbitrary setting)
13 14	O-5(+) O-5(–)	POWER OFF
27 28	O-6(+) O-6(–)	MAINTENANCE

Specifications for contact output are as follows.

Signal type	Input and output circuit	Specifications
Contact signal output		● Contact rating: 125 V AC 0.3 A 30 V DC 1 A ● a contact output

Note

- **Do not apply a load over the maximum rating. Doing so may cause a malfunction.**
- **When switching the load, connect the spark killer, surge absorber (AC or DC load), diode (DC load) etc. in parallel to prevent noise.**
- **“a” contact output: OFF normally (open), ON during operation (short circuit)**

Reference

“ 5.4.1 Signal Allocation ” (page 34)

6.3.2 Contact Input

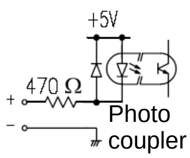
This unit is equipped with contact inputs as shown below. They are enabled when set to the contact point, and the following operations are possible (refer to “ 5.4.2 Signal Input Setting ” (page 36)).

Item	Description	Input timing
NO SAMPLE FLOAT SWITCH	Detects NO SAMPLE using the built-in float switch.	Starts when turned ON (short circuit) for more than 5 seconds after being OFF (open).
TIME CORRECTION	Accepts time corrections.	Accepts time corrections when turned ON (short circuit) for 3 to 10 seconds after being OFF (open).

Allocations for contact input terminals are as follows.

Terminal No.	Output No.	Allocation
7 8	I-1(+) I-1(–)	Float switch
21 22	I-2(+) I-2(–)	Input time correction

Specifications for contact input are as follows.

Signal type	Input and output circuit	Specifications
Contact signal input	 	● “a” contact signal input without voltage (open collector output is possible) ● Insulated input (common to (–) side) ● ON resistance: Max. 100 Ω ● Open voltage: 5.5 V DC ● Short circuit current: Max. 5 mA

Note

- **Do not connect wiring with voltage to the input terminal. Doing so may cause a malfunction.**
- **If the input timing is less than 3 seconds, the contact input may not be accepted. Make sure to allow more than 3 seconds for contact ON time.**

6.4 Serial Input and Output (RS-232C)

This unit includes the serial input and output of RS-232C as standard equipment.
Specifications are as follows.
Regarding details such as command etc, contact our service personnel.

● Transmission format

Baud rate:	19200 bps
Character length:	8 bit
Parity:	N/A
Stop bit:	1 bit
Communication method:	Full-duplex

● Terminal allocation

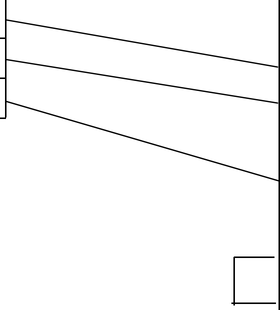
Allocations for terminals are as follows.

Terminal No.	Output No.
1	TXD
2	RXD
3	SG

● Cable specifications

Unit side (3-pin, naked wire)

Name	Pin No.
TXD	1
RXD	2
SG	3



PC side (D-SUB 9-pin female connector)

Pin No.	Name
1	CD
2	RD
3	SD
4	ER
5	SG
6	DR
7	RS
8	CS
9	

6.5 Saving Data to a CF (Compact flash) Card

This unit enables the transfer of saved internal data using a CF card, and the verification of the data on a PC etc. The CF card is available optionally.

Note

- **Make sure to use the CF card produced by HORIBA. We cannot guarantee normal operation when using other products.**
- **The file system of this unit is FAT. FAT32 and NTFS are not accepted.**

How to save to a CF card

Reference

“ 5.8.6 CF Card Transfer ” (page 63)
“ 5.8.7 CF Card Initialization ” (page 64)

Folder configuration to be saved on the CF card

A second folder is created, which is named according to the date and hour of the data transfer, within the first folder where individual ID numbers are named.

Example:

When the individual ID number is "12345678" and the data is transferred at 18 o’clock on October 22, 2004 and 1 o’clock on June 7, 2005:

```
"12345678" folder  ─┬─ "04102218" folder
                    └─ "05060701" folder
```

Data to be saved on the CF card

Saves the data with the following file names in the created folder.

1 hour data history:	HDATA.CSV
1 minute data history:	MDATA.CSV
Alarm history:	ALMDATA.CSV
Zero calibration history:	ZERO.CSV
Span calibration history:	SPAN.CSV
Internal RAM data:	RAMDATA.CSV

The above data files are saved on the CF card simultaneously.

Data Information

Each file is in the CSV (comma separated value) format, and a label line is on the first line. The label meanings are as follows.

(HDATA.CSV)

Label name	Description
YY/MM/DD/HH:MM:SS	year/month/date/hour:minute:second
UV	UV
VIS	VIS
COD	COD
TSS	TSS
UV-VISL	UV-αVIS
DataStat	Data status (determined by each bit) *1
AlmUV	UV data alarm (determined by each bit) *2
AlmVIS	VIS data alarm (determined by each bit) *2
AlmCOD	COD data alarm (determined by each bit) *2
AlmTSS	TSS data alarm (determined by each bit) *2
AlmUVVIS	UV-αVIS data alarm (determined by each bit) *2

The data is in hexadecimal notation. Convert the hexadecimal notation to binary notation according to the conversion table, and determine the values ("1" is ON and "0" is OFF) by referring to each digit (bit) of each table from the left.

Conversion table, hexadecimal notation to binary notation

Hexadecimal notation	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Binary notation	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

*1: Data status

Alarm

Bit0:	invalid data
Bit1:	maintenance
Bit2:	with NO power
Bit3:	spare
Bit4:	calibration
Bit5:	spare
Bit6:	sub, initialization
Bit7:	sampling output hold
Bit8 to 16:	spare
Bit17:	alarm 001 (warm-up)
Bit18:	spare
Bit19:	alarm 003 (cell-motor operation error)
Bit20:	alarm 004 (inside communication error)
Bit21:	alarm 005 (no sample)
Bit22:	alarm 006 (inside communication error (I2C))
Bit23:	alarm 007 (power ON alarm)

Bit24:	alarm 008 (battery cell-motor alarm)
Bit25:	alarm hold 009 (light source malfunction)
Bit26 to 31:	spare

Example) If "DataStat" is "00800006";

1. **Convert the hexadecimal notation to a binary notation.**
   After converting, 0 to 0000, 0 to 0000, 8 to 1000, and so on, "0000/0000/1000/0000/0000/0000/0000/0110" is obtained.
2. **Apply the binary notation value to the bit table from the left.**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Binary notation value	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
Hexadecimal notation value	0				0				8				0				0				0				0				6			

"1" in binary notation is ON. Data status is as follows.

Bit23:	alarm 007 (power ON alarm)
Bit2:	with NO power
Bit1:	maintenance

*2: Data alarm
Determine by conversion using the same method as *1

Bit0:	alarm 0n0 (abnormal upper limit of concentration)
Bit1:	alarm 0n1 (abnormal lower limit of concentration)
Bit2:	spare
Bit3:	spare
Bit4:	spare
Bit5:	alarm 0n5 (zero calibration error)
Bit6:	alarm 0n6 (span calibration error)
Bit7 to 11:	spare
Bit12:	maintenance output data
Bit13:	warm-up output data
Bit14:	spare
Bit15:	alarm output data

(MDATA.CSV)

Same as HDATA.CSV

(ALDATA.CSV)

Label name	Description
YY/MM/DD/HH:MM:SS	year/month/date/hour:minute:second
AlmNo	Alarm No.
DataStat	Data status: ON, OFF

(ZERO.CSV)

[UV]: UV data

[VIS]: VIS data

Label name	Description
YY/MM/DD/HH:MM:SS	year/month/date/hour:minute:second
CalbNo	Calibration No. (M: manual calibration, A: automatic calibration)
CalbStat	Calibration result: OK, NG
CalbConc	Calibration value
CoeffiZero	Zero calibration coefficient
CoeffiSpan	Span calibration coefficient

(SPAN.CSV)

[UV]: UV data

[VIS]: VIS data

[TSS]: TSS data

Note

If the names of files or folders are the same, they shall be overwritten.

(RAMDATA.CSV)

Saves the internal setting table.

(Exclusive for service personnel)

7 Maintenance

Calibration and periodical maintenance are required in order to maintain the normal operation and performance of the unit.
Regarding calibration, refer to “ 4 Calibration ” (page 14).

7.1 Maintenance Item

The maintenance cycle in the table is the typical interval when using under standard conditions, and does not guarantee the measurement values. They may vary according to the circumstances or usage conditions set the proper cycle after using for a certain period of time.

Maintenance item		Maintenance cycle						Time to clean/replace
Subject	Description	Weekly	Monthly	Every half year	Yearly	Every 3 years	Accordingly	
Analyzer unit	Cleaning of measuring cell	✓	✓	✓	✓	✓	✓	When dirt can be seen.
	Inspection and replacement of wipers			✓	✓	✓	✓	When dirt can be seen.
	Replacement of desiccant agent in analyzing case			✓	✓	✓	✓	Periodical replacement (every half year)
	Replacement of desiccant agent in measuring cell			✓	✓	✓	✓	Periodical replacement (every half year)
	Replacement of inner parts				✓	✓	✓	When the measurement value is abnormal
	Replacement of UV lamp				✓	✓	✓	Periodical replacement (yearly) When the measurement value is abnormal When the UV lamp does not illuminate
	Replacement of motor				✓	✓	✓	Periodical replacement (yearly)
	Replacement of seal washers				✓	✓	✓	Periodical replacement (yearly)
	Replacement of V-rings				✓	✓	✓	Periodical replacement (yearly)
	Replacement of gear				✓	✓	✓	Periodical replacement (yearly)
	Replacement of cell				✓	✓	✓	When dirt can be seen When it is damaged
	Replacement of case packing				✓	✓	✓	Periodical replacement (yearly)
Control unit	Replacement of battery for clock					✓	✓	Periodical replacement (every 3 years)
	Replacement of substrate and electric parts						✓	When malfunctioning

Maintenance item		Maintenance cycle						Time to clean/replace
Subject	Description	Weekly	Monthly	Every half year	Yearly	Every 3 years	Accordingly	
Sampling unit	Cleaning of measuring tank	✓	✓	✓	✓	✓	✓	When dirt can be seen When clogged
	Cleaning of overflow tank	✓	✓	✓	✓	✓	✓	When dirt can be seen When clogged
	Cleaning of piping			✓	✓	✓	✓	When dirt can be seen When clogged
	Cleaning of ball valve			✓	✓	✓	✓	When dirt can be seen When clogged

Note

Maintenance items for which the maintenance period is over 1 year are not described in this manual. These should be performed by HORIBA service personnel.

7.2 Maintenance of Analyzer Unit

7.2.1 Cleaning Method for Measuring Cell

Here describes the cleaning method of the measuring cell in the analyzer unit.
Perform maintenance approximately once a week.
Perform this maintenance work with the power ON.

1. **Turn the maintenance switch ON.**
2. **Press the [ACTION] button on the MAINTENANCE screen.**

Reference

Refer to “ 5.6.1 Action ” (page 52).

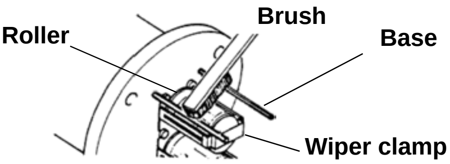
3. **Turn the wiper motor ON and UV lamp OFF.**

4. **Remove the analyzer unit from the measuring tank.**

Note

- **When the wiper motor is operating, take care to handle so that fingers etc are not caught in the motor.**
- **Care should be taken when handling the cell of analyzer unit so that it does not touch other parts of the unit.**
- **Before performing the work, confirm that the UV lamp is not illuminated. The UV lamp should not be viewed directly when it is illuminated as it may damage your eyes.**
- **If no button is pressed for about two hours on the MAINTENANCE - ACTION display, the display automatically turns to the MEASUREMENT display. In this case, care should be taken as the wiper motor and UV lamp may turn ON.**

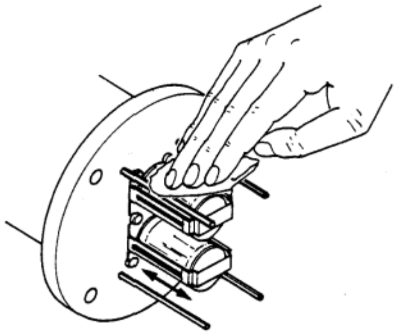
5. **Scrape off the contamination or alga adhered to the peripheral parts such as wiper clamp, base, or roller around the cell.**



Note

- **Do not scrape the cell (glass part) with a brush. The cell may be damaged.**
- **While cleaning, do not tilt the analyzer unit 80 degrees or more from vertical.**
- **Do not leave the analyzer unit standing alone. Put the analyzer unit on the calibration tank when not using the measuring tank, for safe and easy handling.**
- **The measuring cell is made of glass. Handle with great care so as not to break the glass.**
- **Clean the wiper within a few minutes. If operating the wiper without any solution for a long period of time, it may damage the cell.**

6. **Wipe off the rotating cell to remove contamination with a clean, wet cloth by moving it up and down lightly.**



Wipe to remove contamination by moving up and down.

7. **Thoroughly clean the cell and peripheral parts using the cleaning bottle or small amount of tap water.**
   In this case, do not put any solution on the base of cell rotary shaft (V-ring seal part).
8. **Dip the analyzer unit in the calibration tank of zero solution.**
9. **Press the [CLOSE] button of the control unit several times to return to the MEASUREMENT screen.**
10. **Warm up the unit for approximately 60 minutes, replace the zero solution, and perform zero calibration.**

Reference

" 4.4.2 Common Zero Calibration " (page 20).

11. **Perform span calibration as necessary.**

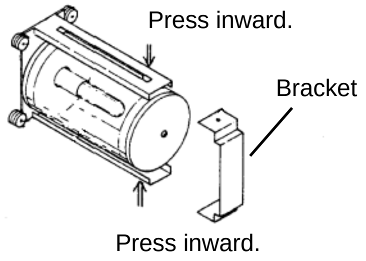
Reference

" 4.4.3 Common Span Calibration " (page 23).

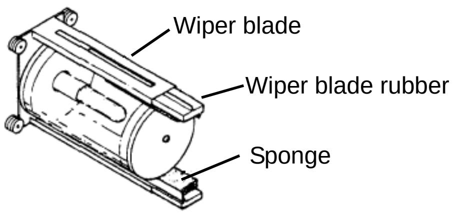
7.2.2 Replacement of Wiper Blade Rubber

Here describes the replacement of wiper blade rubber of measuring cell.

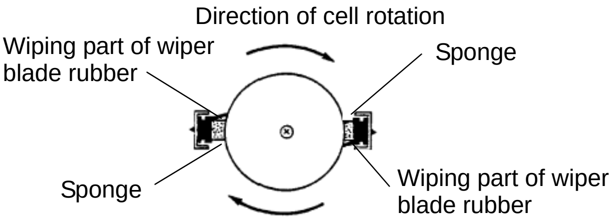
1. **Turn ON the maintenance switch.**
2. **Turn OFF the power switch.**
3. **Remove the analyzer unit from the measuring tank and remove the fixing brackets (2) of the cell unit.**



4. **Pinch the wiper blade rubber with long-nose pliers etc. and remove it by pulling inward and downward.**



5. **Insert the new wiper blade rubber in the reverse order to step 4.**
   Ensure a correct installing direction so that the wiping parts of the wiper blade rubber are located in the direction of cell rotation when viewing from the bottom.



6. **Install the bracket in its original place.**
7. **Put the analyzer unit back in the measuring tank.**
8. **Turn the power switch ON.**
9. **Press the [ACTION] button on the MAINTENANCE screen.**

Reference

“ 5.6.1 Action ” (page 52).

10. Turn the wiper motor ON and UV lamp OFF.

11. Remove the analyzer unit from the measuring tank and confirm that the replaced wiper blade rubber and sponge are in contact with the cell and the wiper blade wiping smoothly. If the wiper blade is misaligned or the rotation is not smooth, repeat wiper blade installation from step 5.

Note

- **As the wiper motor is operating, take care to handle so that fingers etc. are not caught in the motor.**
- **Take care in handling the cell of analyzer unit so that it does not touch other parts of the unit.**
- **Before performing the work, confirm that the UV lamp is not illuminated. The UV lamp should not be viewed directly when it is illuminated as it may damage your eyes.**
- **If no button is pressed for about two hours on the MAINTENANCE - ACTION display, the display automatically turns to the MEASUREMENT display. In this case, care should be taken as the wiper motor and UV lamp may turn ON.**
- **Clean the wiper within a few minutes. If operating the wiper without any solution for a long period of time, it may damage the cell.**

12. Dip the analyzer unit in the calibration tank of zero solution.

13. Warm up the unit for approximately 60 minutes, replace the zero solution, and perform zero calibration.

Reference

“ 4.4.2 Common Zero Calibration ” (page 20).

14. Perform span calibration as necessary.

Reference

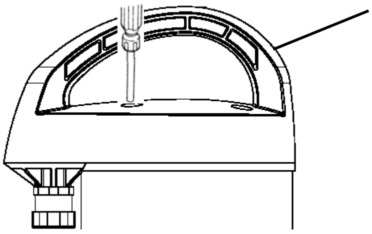
“ 4.4.3 Common Span Calibration ” (page 23).

7.2.3 Replacement of Desiccant Agent and Seal Washers for Screws

Here describes the replacement of desiccant agent and seal washers for screws in the analyzer case.

Replace the desiccant agent and seal washers every 6 months.

1. **Turn ON the maintenance switch.**
2. **Turn OFF the power switch.**
3. **Remove the analyzer unit from the measuring tank and put on the calibration tank.**
4. **Remove the screw covers (4) from the upper part of the analyzer unit and remove the screws (4) using a screwdriver.**



5. **Take the old desiccant agent and replace with the new one attached with this unit (square type, 5 pieces in 1 set). Remove the dirt from the surface of new desiccant agent and put in the analyzer unit with the film surface facing downward.**
6. **Remove the screws and replace the seal washers (4) between the screw and case.**
7. **Install the upper part of analyzer unit, screws (4), and screw covers (4).**

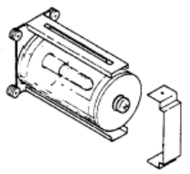
Note

- **Packing for the seal is attached on the circumference of the upper part of analyzer unit. Confirm that the packing is not removed, caught, or contaminated before tightening.**
- **Replaced desiccant agent can be disposed of as a general waste.**

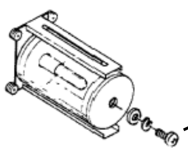
7.2.4 Replacement of Desiccant Agent in the Measuring Cell

Here describes the replacement of desiccant agent in the measuring cell of analyzer unit.
Replace the desiccant agent every 6 months.

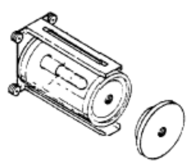
1. **Turn ON the maintenance switch.**
2. **Turn OFF the power switch.**
3. **Remove the analyzer unit from the measuring tank.**
4. **Lightly wipe off moisture adhered to the measuring cell with soft cloth.**
5. **Remove the 2 brackets from the analyzer unit.**
   The following description is the replacement method for the desiccant agent of the light source cell.
   Replace the desiccant agent on the detector side by following the same procedure.



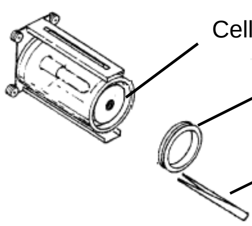
6. **Remove the screw.**



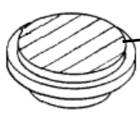
7. **Remove the cell plate D.**



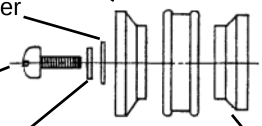
8. **Remove the cell packing by using tweezers.**



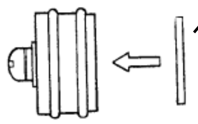
9. **Take the cell plate C and remove the desiccant agent.**



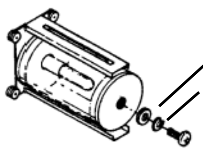
10. **Set the cell plate C, cell packing, and cell plate D as shown in the figure, and tighten the screw for 2 or 3 threads.**



11. **Peel off the backing paper of the two-sided tape of the new desiccant agent, and adhere on the center part of the cell plate C. Before adhesion, brush off the powder on the desiccant agent.**



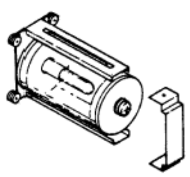
12. **Insert this block into the cell, and fully tighten the screw to attach the cell packing firmly on the inside surface of cell.**



Note

- **Confirm that no moisture or contamination is adhered inside the cell. If there is moisture or contamination, thoroughly wipe it off with a soft cloth.**
- **Tighten the screw after checking that no moisture or contamination is adhered on the seal part of the cell packing.**

13. **Inspect the cell for damage such as breaks or cracks, and set the bracket.**



14. **Replace the desiccant agent on the detector side cell by following the same procedure.**

The replacing procedure of the desiccant agent is completed.

Note

- **Confirm that the cell plate D is installed without any gap between it and the cell plate D. If there is a gap, solution may enter into the cell part leading to malfunction. Therefore, install the cell plate D again.**
- **Replaced desiccant agent can be disposed of as a general waste.**

15. **Put the analyzer unit back in the measuring tank.**

16. **Turn ON the power switch.**

17. **Press the [ACTION] button on the MAINTENANCE screen.**

Reference

“ 5.6.1 Action ” (page 52).

18. **Turn the wiper motor ON and UV lamp OFF.**

19. **Remove the analyzer unit from the measuring tank and confirm that the replaced wiper blade rubber and sponge are in contact with the cell and the wiper blade activate smoothly.**

Note

- **As the wiper motor is operating, take care to handle so that fingers etc are not caught in the motor.**
- **Care should be taken when handling the cell of analyzer unit so that it does not touch other parts of the unit.**
- **Before performing the work, confirm that the UV lamp is not illuminated. The UV lamp should not be viewed directly when it is illuminated as it may damage your eyes.**
- **If no button is pressed for about two hours on the MAINTENANCE - ACTION display, the display automatically turns to the MEASUREMENT display. In this case, care should be taken as the wiper motor and UV lamp may turn ON.**
- **Clean the wiper within a few minutes. If operating the wiper without any solution for a long period of time, it may damage the cell.**

20. **Dip the analyzer unit in the calibration tank with zero solution.**

21. **Warm up the unit for approximately 60 minutes, replace the zero solution, and perform zero calibration.**

Reference

“ 4.4.2 Common Zero Calibration ” (page 20).

22. **Perform the span calibration as necessary.**

Reference

“ 4.4.3 Common Span Calibration ” (page 23).

23. **Put the analyzer unit back in the measuring tank, and turn the maintenance switch OFF.**

7.3 Cleaning of Sampling Unit

Here describes the cleaning method of the sampling unit.
Perform cleaning periodically to remove contamination and sludge etc.

7.3.1 Cleaning of Overflow Tank and Measuring Tank

1. **Turn ON the maintenance switch of the control unit and stop the supply of sample water.**
2. **Turn OFF the power switch.**
   Open the lid of overflow tank. Remove the analyzer unit from the measuring tank.

Note

When removing the analyzer unit from the measuring tank, take care not to damage the cell unit. In addition, put the analyzer unit on the calibration tank with solution (sample water or tap water) so that it does not idle.

3. **Set the valves, (V-1), (V-2), and (V-3) which are on the lower part of overflow tank, and the valve (V-4) on the lower part of measuring tank to fully open. If the supply of sample water cannot be stopped, set the valve (V-1) to close and clean all except V-1.**

Reference

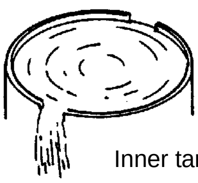
“ 9.2 Piping Flow Chart ” (page 99).

4. **Remove the inner tank of measuring tank.**

Reference

“ 7.3.2 Removal of Inner Tank of Measuring Tank ” (page 89).

5. **Remove the sludge from the overflow tank and measuring tank, and clean the inside with tap water etc. If the flow from the overflow tank to measuring tank is partially clogged, remove the piping of measuring tank and clean the inside of piping with tap water.**
6. **After cleaning the overflow tank, measuring tank, inner tank of measuring tank, and piping, set back to the original condition.**
7. **Set the valves (V-1) and (V-4) to fully close, and (V-2) and (V-3) to fully open.**
8. **Supply the sample water. Open valve (V-1) gradually until sample water runs out of the overflow outlet (1) of the overflow tank. If the sample water does not run out of the overflow outlet (1) even when the valve (V-1) is fully open, then close valve (V-2) gradually to adjust the flow from the overflow outlet (1).**
9. **Ensure that sample water overflows from the upper notches (2) of the inner tank of measuring tank and is discharged from the drain outlet.**



Inner tank of measuring tank

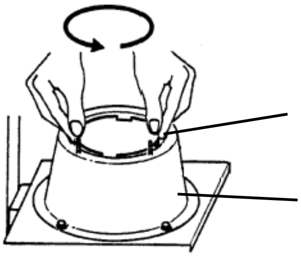
10. **When outlet flow from the overflow tank and the inner tank of measuring tank reaches the designated volume and is running stably, put the lid of overflow tank back on.**

11. **If calibration is required, install the analyzer unit on the calibration tank. If not, install the analyzer unit on the measuring tank.**
12. **Turn the power switch ON.**
13. **After calibration and installation of analyzer unit to the measuring tank, turn ON the maintenance switch when the measurement is stable.**

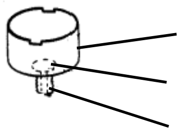
7.3.2 Removal of Inner Tank of Measuring Tank

The removal of the inner tank of the measuring tank is described here.

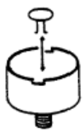
1. **Stop the supply of sample water and turn the power OFF.**
2. **Remove the analyzer unit and blind plate.**
3. **Hold the inner tank of the measuring tank by both hands and turn counterclockwise to remove it.**



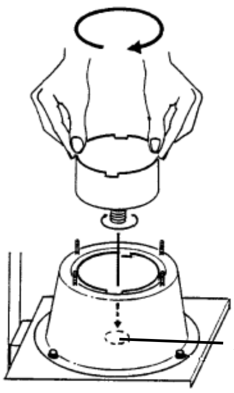
4. **Remove the current plate from the inner tank of the measuring tank. As the spring (leg) of the current plate protrudes out of the screw part on the lower part of the inner tank, it can be removed upward by pressing inside.**



5. **After cleaning the inner tank of measuring tank, put the current plate back in the original position.**



6. **Mount the inner tank to the measuring tank by aligning the screw, and turn clockwise to fix.**



7.4 Accessories and Spare Parts

- **Composition of main body**

OPSA-150 set

Items

Control unit
Analyzer unit
Measuring tank
Overflow tank
Accessories

- **Accessories**

Part name	Qty.	Specification
Calibration solution H	1 box	For 1.0 Abs (6 bottles/1 set)
Calibration tank	2	Calibration tank for OPSA
Wiper blade rubber	1 set	For cell wiper of OPSA (4 pieces/1 set)
Desiccant agent	1 set	For analyzer unit case (5 pieces/1 set)
Desiccant agent	1 set	For analyzer unit measuring cell (2 pieces/1 set)
Flat-blade screwdriver	1	For tightening the screw
Seal washer	1 set	For analyzer unit upper case (4 pieces/1 set)
Instruction manual (this manual)	1	OPSA-150

- **Spare parts**

Part name	Qty.	Specification	Spare parts No.
CF card	1	HORIBA spec.	3014030160
Wiper blade rubber	1 set	For measuring cell wiper (4 pieces/1 set)	3200043531
Desiccant agent	1 set	For analyzer unit case (5 pieces/1 set)	3200044316
Desiccant agent	1 set	For analyzer unit measuring cell (2 pieces/1 set)	3200044334
Calibration solution H	1	For 1.0 Abs (6 bottles/1 set)	3100118964
Calibration solution H II	1	For 2.0 Abs (6 bottles/1 set)	3100118965
Seal washer	1 set	For analyzer unit upper case (4 pieces/1 set)	3200043872

8 Troubleshooting

8.1 Table of Status and Operation

Item	Status	LED (POWER/ALARM) ◎ : green light on ○ : green blinking	Contact (batch alarm)	Contact (maintenance)	Output status	Alarm display	Alarm history
Common	MEAS.	◎			Instantaneous value output		
	MAINTENANCE	○			Maintenance output	○	Maintenance
	CALIBLATION	○			Maintenance output	○	Maintenance
	DATA	◎			Instantaneous value output		

8.2 Table of alarm and Operation

Item	Status	LED (POWER/ALARM) ◎ : green light on ● : red light on ○ : green blinking △ : red blinking	Contact (batch alarm)	Contact (analyzer alarm)	Output status	Alarm display	Alarm history
Common	WARM-UP	○			Warm-up output	○	○
	CELL MOTOR ERR.	●	○		Alarm output	○	○
	SENSOR COMM.ERR.*1	●	○	○	Alarm output	○	○
	SAMPLE LACK	△			Instantaneous value output	○	○
	PIO COMM.ERR.*2	●	○	○	Alarm output	○	○
	POWER ON/OFF				Instantaneous value output		○
	BATTERY ERR.	△			Instantaneous value output	○	○
	LAMP ERR.	●	○		Alarm output	○	○
UV	ZERO CAL.	△			Instantaneous value output	○	○
	SPAN CAL.	△			Instantaneous value output	○	○
VIS	ZERO CAL.	△			Instantaneous value output	○	○
	SPAN CAL.	△			Instantaneous value output	○	○
COD	HIGH	△			Instantaneous value output	○	○
	LOW	△			Instantaneous value output	○	○

Item	Status	LED (POWER/ALARM) ◎ : green light on ● : red light on ○ : green blinking △ : red blinking	Contact (batch alarm)	Contact (analyzer alarm)	Output status	Alarm display	Alarm history
TSS	HIGH	△			Instantaneous value output	○	○
	LOW	△			Instantaneous value output	○	○
	SPAN CAL.	△			Instantaneous value output	○	○

*1 SENSOR COMM.ERR: Communication error between analyzer unit substrate and control unit substrate
*2 PIO COMM.ERR: Communication error between control unit CPU substrate and input/output substrate

8.3 Alarm Occurrence Condition

Item	Alarm	Alarm triggered	Alarm cleared	Alarm release *3
Common	WARM-UP	For 60 minutes after power ON	After 60 minutes passed since power ON	○
	CELL MOTOR ERR.	When a rotation of the motor takes 5 seconds or more.	When a rotation of motor takes less than 5 seconds.	△
	SENSOR COMM. ERR. *1	When analyzer unit communication is abnormal.	When analyzer unit communication resumes.	△
	SAMPLE LACK	When contact input of no sample is ON for 5 seconds or more.	When contact input of no sample is OFF for 5 seconds or more.	△
	PIO COMM. ERR. *2	When communication of the input/ output substrate is abnormal	When communication of the input/ output substrate resumes.	△
	POWER ON/OFF	History is saved at start up.		
	BATTERY ERR.	When the backup battery voltage for clock is less than 2.4 V.	When the backup battery voltage for clock is 2.4 V or over.	○
	LAMP ERR.	When the light intensity of the light source is decreased.	When the light intensity of the light source is restored.	△
UV	ZERO CAL.	When zero calibration value is out of the calibration range.	When zero calibration value is within the calibration range.	○
	SPAN CAL.	When span calibration value is out of the calibration range.	When span calibration value is within the calibration range.	○
VIS	ZERO CAL.	When zero calibration value is out of the calibration range.	When zero calibration value is within the calibration range.	○
	SPAN CAL.	When span calibration value is out of the calibration range.	When span calibration value is within the calibration range.	○
COD	HIGH	The status "measurement value > MAX ALARM VALUE" continues longer than the specified alarm time.	The status "measurement value ≤ MAX ALARM VALUE" continues longer than the specified alarm time.	△
	LOW	The status "measurement value < MIN ALARM VALUE" continues longer than the specified alarm time.	The status "measurement value ≥ MIN ALARM VALUE" continues longer than the specified alarm time.	△
TSS	HIGH	The status "measurement value > MAX ALARM VALUE" continues longer than the specified alarm time.	The status "measurement value ≤ MAX ALARM VALUE continues while exceeding the specified alarm time.	△
	LOW	The status "measurement value < MIN ALARM VALUE" continues longer than the specified alarm time.	The status "measurement value ≥ MIN ALARM VALUE" continues longer than the specified alarm time.	△
	SPAN CAL.	When span calibration value is out of the calibration range.	When span calibration value is within the calibration range.	○

*1 SENSOR COMM.ERR.: Communication error between analyzer unit substrate and control unit substrate

*2 PIO COMM.ERR.: Communication error between control unit CPU substrate and input/output substrate

*3 Alarm release: Forced stop by pressing [DELETE] on the ALARM screen.

○ : Forced stop is possible.

△ : Though forced stop is possible, the alarm recurs because the cause has not been resolved.

8.4 Causes and Measures for Alarm

Item	Alarm	Causes	Measures	Refer
Common	WARM-UP	Waiting to be stabilized after the analyzer unit power ON.	No problem. (It takes approximately 60 minutes until the analyzer unit is stabilized.)	" 3.2 Starting Operation " (page 12)
	CELL MOTOR ERR.	Contamination of measuring cell.	Clean the measuring cell.	" 7.2.1 Cleaning Method for Measuring Cell " (page 80)
		Failure in analyzer unit cell-motor.	Replace the motor.	Service call
	SENSOR COMM. ERR. *	Poor connection of analyzer unit cables.	Connect the analyzer unit cables properly.	
		Poor connection of analyzer unit cables or control unit, or faulty parts.	Connect the inside cables of the analyzer unit and control unit again.	Service call
	SAMPLE LACK	Sample water supply is stopped.	Check the sample water supply and turn supply on.	" 3.1 Preparation for Operation " (page 11)
		Valve shut in improper condition.	Adjust the valves V1 to V4 in proper condition.	" 3.1 Preparation for Operation " (page 11)
		Clogging of piping to the overflow tank.	Clean the piping.	" 7.3 Cleaning of Sampling Unit " (page 88)
		Failure in float switch for no sample detection.	Replace the float switch.	Service call
	PIO COMM. ERR. *	Communication error between input/output substrate and CPU substrate of control unit.	Turn the power OFF/ON. This may solve the problem.	
		Poor connection of control unit cables, or faulty parts.	Connect the inside cables of control unit again.	Service call
	POWER ON/OFF	Power OFF by the electric outage.	No problem.	
	BATTERY ERR.	Backup battery for clock has expired.	Replace the clock backup battery.	Service call
	LAMP ERR.	Contamination of measuring cell.	Clean the measuring cell.	" 7.2.1 Cleaning Method for Measuring Cell " (page 80)
		Light source is temporarily off.	Lamp may not be illuminated until the light source is warmed up. Turn the power ON/OFF and wait for approximately 30 minutes as it may be restored.	Service call
		Output is decreased by deterioration of light source.	Replace the light source.	Service call
		Poor connection of control unit cables, or faulty detector or substrate	Connect the inside cables of control unit again.	

Item	Alarm	Causes	Measures	Refer
UV	ZERO CAL.	Incompatible calibration solution.	Make sure to use distilled water for the calibration solution.	
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	" 7.2.1 Cleaning Method for Measuring Cell " (page 80) " 7.2.2 Replacement of Wiper Blade Rubber " (page 82)
		Failure of parts in the analyzer unit.	Replace the faulty parts.	Service call
	SPAN CAL.	Incompatible calibration solution.	Make sure to use the exclusive calibration solution, and distilled water as the dilution water.	" 4.4.3 Common Span Calibration " (page 23)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	" 7.2.1 Cleaning Method for Measuring Cell " (page 80) " 7.2.2 Replacement of Wiper Blade Rubber " (page 82)
		Failure of analyzer unit parts.	Replace the faulty parts.	Service call
VIS	ZERO CAL.	Incompatible calibration solution.	Make sure to use distilled water for the calibration solution.	
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	" 7.2.1 Cleaning Method for Measuring Cell " (page 80) " 7.2.2 Replacement of Wiper Blade Rubber " (page 82)
		Failure of analyzer unit parts.	Replace the faulty parts.	Service call
	SPAN CAL.	Incompatible calibration solution.	Check the calibration solution.	" 4.4.3 Common Span Calibration " (page 23)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	" 7.2.1 Cleaning Method for Measuring Cell " (page 80) " 7.2.2 Replacement of Wiper Blade Rubber " (page 82)
		Failure of analyzer unit parts.	Replace the faulty parts.	Service call

Item	Alarm	Causes	Measures	Refer
COD	HIGH	Unsuitable upper limit of setting value.	Review the upper limit of the setting value.	“ 5.5.6 Setting of MAX Alarm Value for COD and TSS ” (page 50)
		Abnormal upper limit of sample water.	Check the sample water.	
		Incompatible calibration solution.	Perform zero and span calibration.	“ 4 Calibration ” (page 14)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	“ 7.2.1 Cleaning Method for Measuring Cell ” (page 80) “ 7.2.2 Replacement of Wiper Blade Rubber ” (page 82)
	LOW	Unsuitable lower limit of setting value.	Review the lower limit of the setting value	“ 5.5.7 Setting of MIN Alarm Value for COD and TSS ” (page 51)
		Abnormal upper limit of sample water.	Check the sample water.	
		Incompatible calibration solution.	Perform zero and span calibration.	“ 4 Calibration ” (page 14)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	“ 7.2.1 Cleaning Method for Measuring Cell ” (page 80) “ 7.2.2 Replacement of Wiper Blade Rubber ” (page 82)
TSS	HIGH	Unsuitable upper limit of setting value.	Review the upper limit of the setting value.	“ 5.5.6 Setting of MAX Alarm Value for COD and TSS ” (page 50)
		Abnormal upper limit of sample water.	Check the sample water.	
		Incompatible calibration solution.	Perform zero and span calibration.	“ 4 Calibration ” (page 14)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	“ 7.2.1 Cleaning Method for Measuring Cell ” (page 80) “ 7.2.2 Replacement of Wiper Blade Rubber ” (page 82)
	LOW	Unsuitable lower limit of setting value.	Review the lower limit of setting value.	“ 5.5.7 Setting of MIN Alarm Value for COD and TSS ” (page 51)
		Abnormal upper limit of sample water.	Check the sample water.	
		Incompatible calibration solution.	Perform zero and span calibration.	“ 4 Calibration ” (page 14)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	“ 7.2.1 Cleaning Method for Measuring Cell ” (page 80) “ 7.2.2 Replacement of Wiper Blade Rubber ” (page 82)
	SPAN CAL.	Unsuitable calibration solution	Check the calibration solution.	“ 4.4.3 Common Span Calibration ” (page 23)
		Contamination of measuring cell.	● Clean the measuring cell. ● Replace the wiper blade rubber when it is deteriorated.	“ 7.2.1 Cleaning Method for Measuring Cell ” (page 80) “ 7.2.2 Replacement of Wiper Blade Rubber ” (page 82)
		Failure of analyzer unit parts.	Replace the faulty parts.	Service call

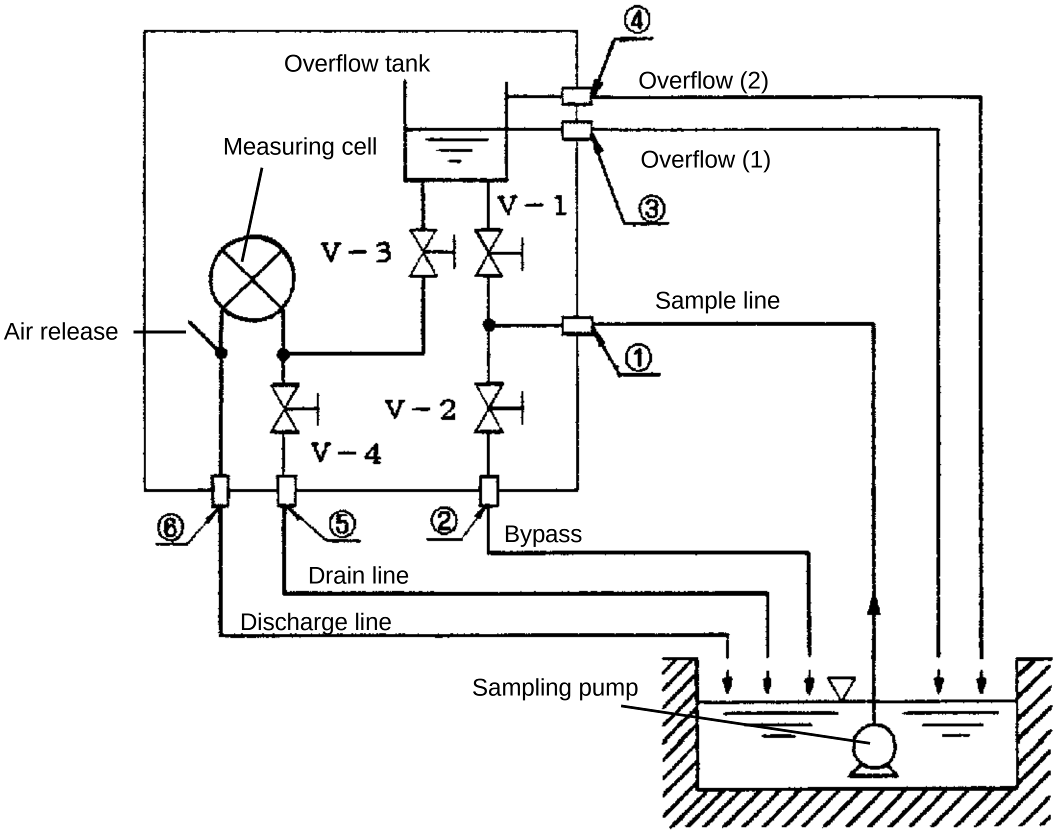
9 Reference Data

9.1 Specifications

Product name	Organic Pollutant Monitor (UV meter)
Model	OPSA-150 (For Vietnam)
Measurement item	UV absorbance, VIS absorbance, COD conversion concentration, TSS conversion concentration
Measurement principle	Dual optical path, Dual wavelength, rotary cell length modulation method
Measurement wavelength	Ultraviolet: 253.7 nm, Visible light: 546.1 nm
Structure of analyzer unit	Circulation type
Measurement range (converted by cell 10mm)	UV absorbance/VIS absorbance Full scale: 0.1 Abs to 5.0 Abs (setting by 0.1 Abs is possible)
Minimum scale	0.0001 Abs (changeable to 0.001 Abs)
Repeatability	Within ±2.0% of full scale (Within ±5.0% of full scale for 2.6 Abs to 5.0 Abs)
Linearity	Within ±2.0% of full scale (Within ±5.0% of full scale for 2.6 Abs to 5.0 Abs)
Stability	Within ±2.0%/24 h of full scale (Within ±4.0%/24 h of full scale for 2.6 Abs to 5.0 Abs)
Response time	T90, within 1 minute (flow rate of approximately 5 L/min)
Cleaning method (cleaning cycle)	Automatic cleaning by wiper (automatic continuous cleaning)
Display method	LCD display: 320 × 240 monochrome graphics LCD data display with dot backlighting (a touch-panel) Display content: UV absorbance, VIS absorbance, UV-VIS absorbance, COD conversion concentration, TSS conversion concentration
Calibration method	Solution calibration using zero solution and span solution (calibration solution in the ampoule) (one-touch calibration is possible)
Sample water condition	Temperature: 2°C to 40°C (should not be frozen) Flow rate of sample water: Min. 2 L/min, Max. 20 L/min * Keep the unit warm (or preserve the heat) to ensure protection against frozen sample water in the cold weather.
Ambient temperature and humidity	Ambient temperature: 0°C to 40°C, Ambient humidity: 85% or below
Analog output	Point: 3 Type: arbitrary 3 items from UV absorbance, VIS absorbance, UV-VIS absorbance, COD conversion concentration, and TSS conversion concentration (selectable) Specification: 4 mA to 20 mA DC or 0 mA to 16 mA DC, insulated output (however, non-insulated among each channel) Maximum load resistance: 600 Ω

Contact output	Point: 6 Type: arbitrary 4 items from POWER OFF, MAINTENANCE (standard, fixed allocation) BATCH ALARM, COD MAX ALARM, TSS MAX ALARM (standard), LIGHT SOURCE MALFUNCTION, NO SAMPLE, CLEANING MOTOR ERROR, ANALYZER ERROR (selectable) Content: POWER OFF: when the power OFF occurred. BATCH ALARM: CLEANING MOTOR ERROR, LIGHT SOURCE MALFUNCTION, ANALYZER ERROR MAINTENANCE: at in-maintenance mode and calibration mode, and when maintenance switch is ON. Specification: contact output without voltage, a contact Contact rating: 125 V AC/0.3 A, 30 V DC/1 A (however, for load resistance) each output COM-independent
Contact input	Point: 2 Type: Float switch input, time adjustment input Specification: contact input without voltage (open collector can be connected), insulated input ON resistance: Max. 100 Ω, Open voltage: 5.5 V DC, Short circuit current: Max. 10 mA
Communication	Interface: RS-232C compliant Communication speed: 19200 bps
Data memory	Memorize the data of measurement value of the measurement item in the main body. Memorized data can be transferred to the CF card. Memory interval: 1 minute or 1 hour Memory time: on the hour every hour Data memory time: interval of 1 minute: approximately 10 days interval of 1 hour: approximately 1 year * The latest data is memorized.
Light source/detector	Light source: low pressure mercury-vapor lamp, detector: silicon photo diode
Piping sockets	● Sample inlet Rp-1/2 female ● Bypass outlet Rc-1/2 female ● Overflow outlet (1) Elbow fitting, 13A (nominal) ● Overflow outlet (2) Elbow fitting, 20A (nominal) ● Drain outlet Rc-1/2 female ● Discharge Tube coupling, 50A (nominal)
Structure	Weather-proofed for outdoor installation (equivalent to IP54)
Main material of each unit	SUS, PVC, PP, CR, SiO_2
Power	100 V to 230 V AC ±10%, 50/60 Hz
Power consumption	100 V to 120 V AC, Max. 45 VA 200 V to 230 V AC, Max. 60 VA
Mass	Control unit: approximately 5.0 kg Analyzer unit: approximately 5.6 kg
Dimension	Control unit: 240 (W) × 104 (D) × 320 (H) mm Analyzer unit: 200 (W) × 180 (D) × 403 (H) mm (Protruding portion are not included.)
Painting color	Control unit: Munsel 5PB 8/1
Conditions for installation	● Flat and stable places free from strong vibration or shock ● Places free from dirt or dust, or where no corrosive gases will be generated. ● At the atmospheric pressure ● Places not subject to direct sunlight ● Places well-ventilated ● Altitude of 2000 m or below

9.2 Piping Flow Chart

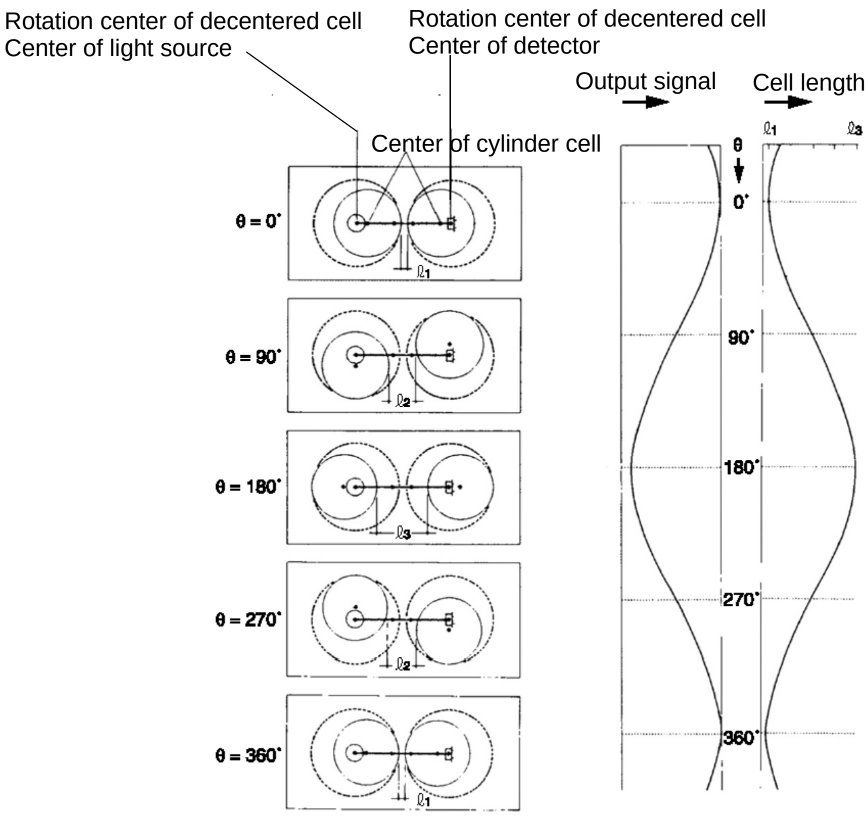


①	Sample inlet	Rp-1/2 female
②	Bypass outlet	Rc-1/2 female
③	Overflow outlet (1)	Elbow fitting, 13A (nominal)
④	Overflow outlet (2)	Elbow fitting, 20A (nominal)
⑤	Drain outlet	Rc-1/2 female
⑥	Discharge	Tube coupling, 50A (nominal)

9.3 Measurement Principle

The rotary cell length modulation method, which combines high performance and reliability, is adopted in this unit.

Two cylinder cells are placed, arranged with the light source and detector in a position where they are decentered from the center of cylinder as shown in the figure below. By rotating these two cylinder cells with the centers of the light source and detector as the rotation center, the distance between the 2 cylinders (cell length) is modulated continuously and periodically. The signal input to the detector from the light source is the signal which changes the difference in light intensity between the maximum cell length (θ = 180 degrees) and the minimum cell length (θ = 0 degree) into the amplitude. Wide range measurement becomes possible by selecting the optimum cell length and calculating according to the change of cell length.



● **Optical compensation**

Optical compensation is attained by the rotary cell length modulation method. Information required for optical compensation is the change of light intensity due to the light source, detector, contamination of the measuring cell etc. This change in light intensity information can be obtained by monitoring the light intensity when at the minimum cell length (θ = 0 degree).

● **TSS compensation**

This unit can measure ultraviolet (UV) and visible light (VIS) absorbances independently. UV-αVIS display is possible by setting the TSS correction factor (α), thus a more reliable TSS compensation can be performed.

9.4 TSS Measurement

TSS measurement is performed by using the visible light (VIS) absorbance.

The measurement method is the transmitting absorption method.

Measure in the measurement range of 200 or below. If measuring in a measurement range that exceeds the specified value, the linearity deteriorates due to the scattering light influence.

9.5 How to Calculate the COD Conversion Factor

If it is judged that there is obviously a linear function between the measurement concentration Xi (i = 1, 2, ···n) and corresponding manual analysis value Y i (i = 1, 2, ···n), the calibration curve formula (regression formula) is expressed as follows.

$$Y = a + bX$$

The gradient of this regression line X-axis (b) and the Y-axis segment (b) can be calculated by using the measurement value and the least-square method as in the following formula.

$$a = \overline{Y} - b\overline{X}$$

$$b = \frac{\Sigma(X_i - \overline{X})(Y_i - \overline{Y})}{\Sigma(X_i - \overline{X})^2} = \frac{n\,\Sigma X_i Y_i - (\Sigma X_i)(\Sigma Y_i)}{n\,\Sigma X_i^{\,2} - (\Sigma X_i)^2}$$

“a” and “b” are rounded to a value which is larger by 1 digit according to the number of significant figures of the measurement value.

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